



GPL813P01X Programming Guide

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Revision History

Revision	Date	By	Remark
V1.4	09/14/2023	Ken Liu	1. Add the difference in standby or Halt mode wake-up CPU stabilization time between GPL813V00A and GPL813P01X in chapter 2.1 Introduction. 2. Add the difference between LVD interrupts in chapter 1.3 Differences between GPL813V00A, GPL813P01X.
V1.3	07/11/2023	Ken Liu	1. Add LCD DPRAM description in chapter 1.3 Differences between GPL813V00A, GPL813P01X. 2. Modify register default value is random in chapter 6.3/6.4/6.7. 3. Modify sleep sample code in chapter 9.4 Procedure to Sleep
V1.2	06/27/2019	Ken Liu	1. Add GPL813P01X body
V1.1	10/15/2018	Ken Liu	1. Add LCD Green mode description in chapter 1.3 Differences between GPL813V00A, GPL813P01X. 2. Add Crystal 32768Hz Capacitance Default description in chapter 1.3 Differences between GPL813V00A, GPL813P01X.
V1.0	05/30/2018	Ken Liu	Original

1 INTRODUCTION

1.1 General Description

GPL813P01X, a special designed CMOS 8-bit microprocessor by Generalplus, offers the best cost/performance ratio in the industry. It features RAM, OTP ROM, I/Os, an interrupt controller, and an automatic display controller/driver in a small device. The extraordinary features are capable of operating in low voltage range from 2.0V ~ 3.6V / 2.0~5.5V. This device is suitable for many application fields such as low power Thermometer, Hygrometer, watch and other LCD related products.

1.2 Features

1.2.1 GPL813P01X

- Built-in 8-bit high performance processor
- Built-in internal regulator for system operation
- 256 byte SRAM
- 8K-byte OTP ROM
- 32-byte DPRAM
- 4MHz RC oscillator for system operation
 - Adjustable CPU clock speed: 1, 1/2, 1/4, 1/8, 1/16 for Rosc
- 32768Hz $\pm 5\%$ RC oscillator and 32768Hz crystal oscillator circuit for time-base.
- Low operating voltage: 2.0V ~ 3.6 V @ -20 ~ 70°C, Vreg=VDD.
2.0V ~ 5.5V @ -20 ~ 70°C, Vreg=Regulator on(2.7V).
- Low standby current, ISTBY < 3uA @3.0V@25°C
- 36 general I/O pins.
 - PA[3:0] (shared with SEG[18:21])
 - PB[7:0] (shared with COM[0:7])
 - PC[3:0] (shared with SEG[10:13])
 - PC[5:4] (shared with [X32O,X32I])
 - PC[7:6] (shared with SEG[8:9])
 - PD[7] (shared with SEG[26])
 - PD[6] (PD6 can be used as EXT2INT)
 - PD[5:3] (shared with SEG[27:29])
 - PD[2] (PD2 can be used as EXTINT)
 - PD[1:0] (shared with SEG[30:31])
 - PE[7:4] (shared with SEG[14:17])
 - PE[3:0] (shared with SEG[22:25])
 - Built-in RFC function (PD2 as input, PD[5:3] as output)
- Internal regulator for LCD operation, 4-level contrast control(2.8V/3.0V/3.3V/VDD)
- LCD configurations: 8 coms x 32 segs (MAX), 6X32, 5X32, 4X32, 3X32
- LCD 1/2, 1/3, 1/4, bias, and 1/3, 1/4, 1/5, 1/6, 1/8 duty
- One 16-bit reloadable timer/counter (Timer0), one 8-bit reloadable timer/counter (Timer1)
- 4-channel PWMIO Output
 - PWM output pin shared with PA[3:0]

- Watchdog mode
- Nine interrupt sources: 2KHz, 128Hz, TMBB, TMBA, EXT1(PD2), EXT2(PD6), Timer0, Timer1, LVD.
- Power down mode: Wake-up sources: key change, TMBB, TMBA, 128Hz, Timer0, and Timer1.
- 2.2V/2.4V/2.7V/3.3V low voltage detector
- Low voltage reset to avoid system runaway from abnormal operation at low voltage.

1.3 Differences between GPL813V00A, GPL813P01X

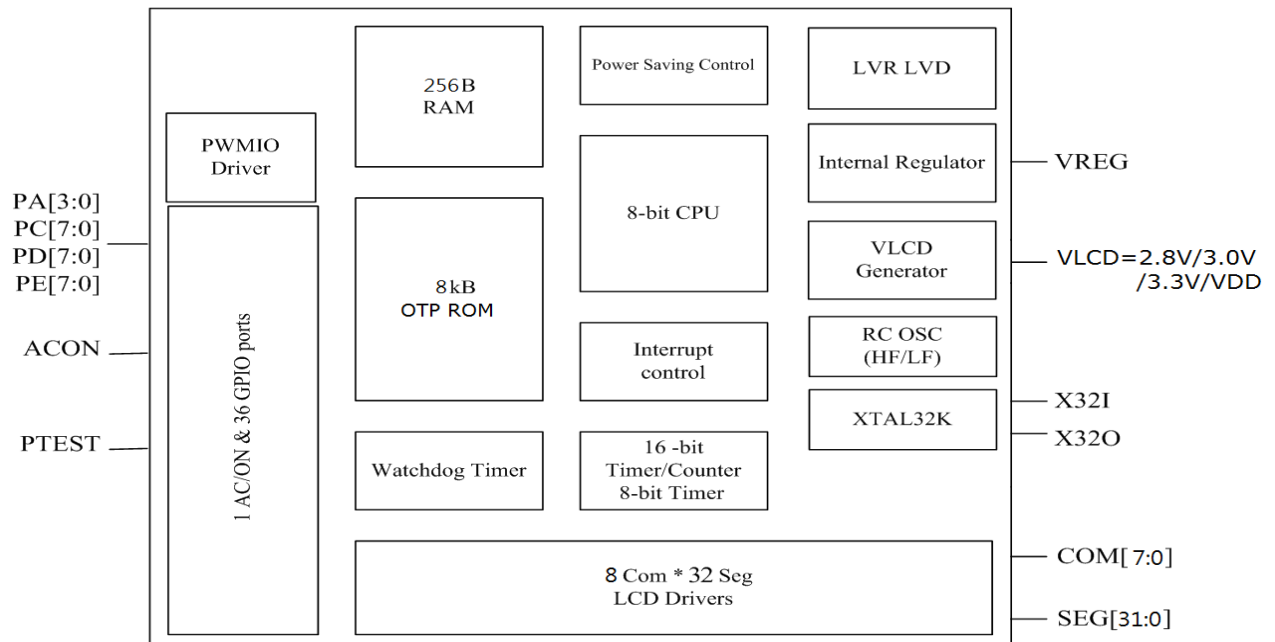
	GPL813V00A (EV Chip)	GPL813P01X
VDD	3V	3V/5.5V
ROM Size	8K bytes	8K bytes
LCD RAM	128 bytes The following addresses are valid: \$3E00 - \$3E7F. If illegal reset is enabled, attempting to operate on these addresses will not trigger an illegal reset.	32 bytes The following addresses are valid: \$3E00-\$3E03, \$3E10-\$3E13, \$3E20-\$3E23, \$3E30-\$3E33, \$3E40-\$3E43, \$3E50-\$3E53, \$3E60-\$3E63, and \$3E70-\$3E73. However, the addresses \$3E04-\$3E0F, \$3E14-\$3E1F, \$3E24-\$3E2F, \$3E34-\$3E3F, \$3E44-\$3E4F, \$3E54-\$3E5F, \$3E64-\$3E6F, and \$3E74-\$3E7F are invalid. If illegal reset is enabled, attempting to operate on these addresses will result in an illegal reset.
LCD Resolution	8 COM x 32 SEG	8 COM x 32 SEG
LCD Duty	1/3, 1/4, 1/5, 1/6, 1/8	1/3, 1/4, 1/5, 1/6, 1/8
I/O Number	PA[3:0] & SEG[18:21] Independent	PA[3:0] shared with SEG[18:21]
	PB[7:0] & COM[0:7] Independent	PB[7:0] shared with COM[0:7]
	PC[7:6] & SEG[8:9] PC[3:0] & SEG[10:13] PC[5:4] & X32K[XTO,XTI] Independent	PC[7:6] shared with SEG[8:9] PC[3:0] shared with SEG[10:13] PC[5:4] shared with X32K[XTO,XTI]
	PD[7] & SEG[26] PD[5:3] & SEG[27:29] PD[1:0] & SEG[30:31] Independent	PD[7] shared with SEG[26] PD[5:3] shared with SEG[27:29] PD[1:0] shared with SEG[30:31]
	PE[7:4] & SEG[14:17] PE[3:0] & SEG[22:25] Independent	PE[7:4] shared with SEG[14:17] PE[3:0] shared with SEG[22:25]

	GPL813V00A	GPL813P01X
Build-in RC Oscillator	CPU clock = 225K / 500KHz/ 1000K/ 1800K/ 2500K/ 3000K /4000KHz	CPU clock = 4000KHz @Vreg =VDD/Regulator(2.7V)

	GPL813V00A	GPL813P01X
	@Vreg = 1.5V only.	
Operating Voltage	VDD = 1.8 ~ 3.6V VDD = 2.4 ~ 5.0V	VDD = 2.0 ~ 3.6V VDD = 2.0 ~ 5.5V
LCD VLCD	Pump capacitor VLCD Set \$0019, P_LCD_VLCD_Ctrl [3:0]	No capacitors VLCD Set \$0019, P_LCD_VLCD_Ctrl [1:0]
LCD Green mode	None	\$0019, P_LCD_VLCD_Ctrl [5:4] Must Write "10" or "11" to enter halt green mode, will make Halt current less.
Crystal 32768Hz Capacitance Default.	P_CLK_32768_En[0]: Default : 1 With External CAP. Note: XTI/XTO needs External Capacitance with X'tal32K.	P_CLK_32768_En[0]: Default : 1 With Internal CAP. Note: GPL813P01X Enable XTI/XTO with Internal Capacitance.
Pull-low Resistor	160kOhm	55kOhm
Function: TimerBase Wakeup	GPL813V00A supported TimerBase Wakeup : 4096Hz P_TIMER_TimeBase_Sel [2:0] "111"	GPL813P01X not supported TimerBase Wakeup : 4096Hz P_TIMER_TimeBase_Sel [2:0] "111"
Function: LVD Interrupt	GPL813V00A LVD Interrupt can be set enabled in Normal and halt mode.	GPL813P01X LVD Interrupt can be set enabled in Normal Mode, but LVD Interrupt must set disabled when entering halt mode. Note: When entering the halt mode, LVD Interrupt must be disabled. When waking up from halt mode, set LVD Interrupt enabled to detect LVD level.

1.4 Block Diagram

1.4.1 GPL813P01X Block Diagram



1.5 Application Fields

- Humidity/temperature multi-meter
- Watch/Alarm clock

2 POWER ON/WAKE UP STABLE TIME

2.1 Introduction

When EV chip GPL813V00A powers on or wakes up from standby mode (32K clock off, CPU clock off), CPU will **not** execute the program immediately. It will first detect LCD's VLCD voltage to assure the VLCD voltage is stable before executing program. Waiting for VLCD to become stable requires a certain period of time that user should be very careful the effect to the system. For example, under the application when key-change wakes up from standby mode, if key is pressed and released rapidly, CPU will not be able to execute the key-scan program after being awakened. As a result, CPU is unable to identify which key has been pressed.

LCD's VLCD will pump the V1x to VLCD via a pumping circuit. In addition, the clock source of the pumping function comes from 32KHz clock. Under the halt mode (32K clock on & CPU clock off), pumping function remains working as well as VLCD voltage remains stable. Therefore, CPU will immediately execute the program after wakeup.

When real chip GPL813P01X powers on or wakes up from standby mode (32K clock off, CPU clock off), CPU will execute the program immediately. It will first start-up CPU clock and then start-up X32K clock waiting for stable. When X32K clock is stable, LCD's VLCD will output immediately.

Start-up CPU clock immediately	GPL813V00A	GPL813P01X
Power on/ Stand-by mode wake up	NO	YES
Halt mode wake up	YES	YES

Note: When Power on/ Stand-by mode wake up, EV chip GPL813V00A needs waiting for VLCD stable time, then Start-up CPU clock.

The difference in standby or Halt mode wake-up CPU stabilization time between GPL813V00A and GPL813P01X.

Key Wake up CPU stable time	GPL813V00A	GPL813P01X
R32K-Stand-by	68ms	530us
R32K-Halt mode	108us	530us
X32K-Stand-by	304ms	530us
X32K-Halt mode	108us	530us

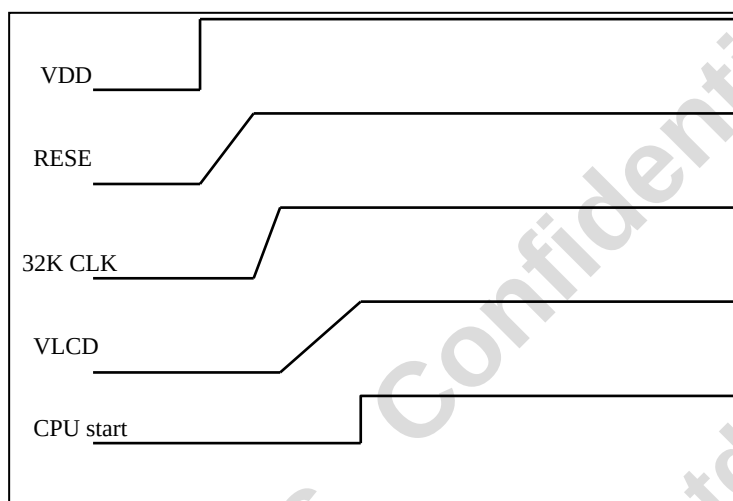
Note: GPL813V00A CPU stable time include 32K Start-up and VLCD stable time

2.2 Timing Sequence of Power on/Wake up

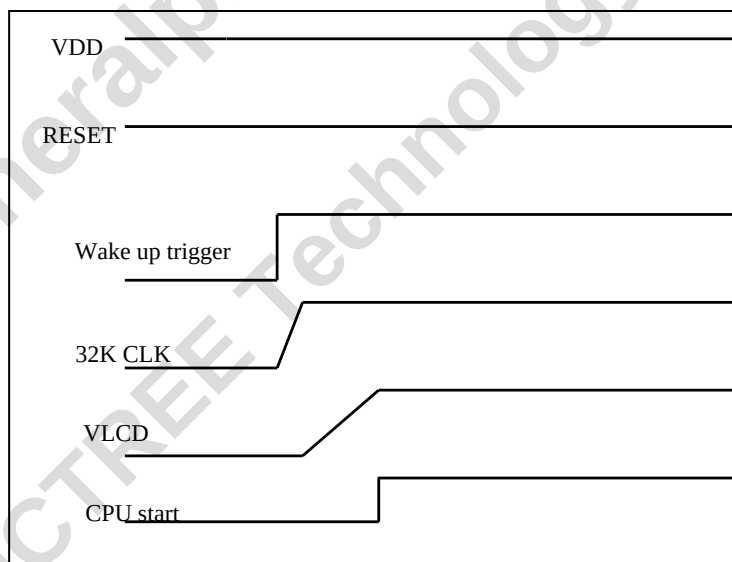
The following diagrams depict the signal sequence of power on and standby mode wakeup. The signal of reset pin is the same as that of power on, in which, 32KHz clock stabilization time may vary significantly based on which OSC is used, R32K or crystal 32K(code option). When Crystal 32K is used, it requires a long time to oscillate. It depends on the capacitor used for Crystal 32KHz. The VLCD stabilization time can refer to the following table for your reference. The actual values given here may vary due to the differences between systems.

GPL813V00A – Power ON/Stand-by Wakeup VLCD stable time						
Condition: 1/4 bias, corresponding capacitor value						
CUP pin	V1 pin	V2 pin	V3 pin	VLCD pin	Vreg pin	Stable time
0.1uF	0.1uF	0.1uF	0.1uF	0.1uF	0.1uF	56ms
1uF	1uF	1uF	1uF	1uF	0.22uF	80ms

Note: When Power on or Stand-by wakeup, VLCD stable time will corresponding capacitor value.



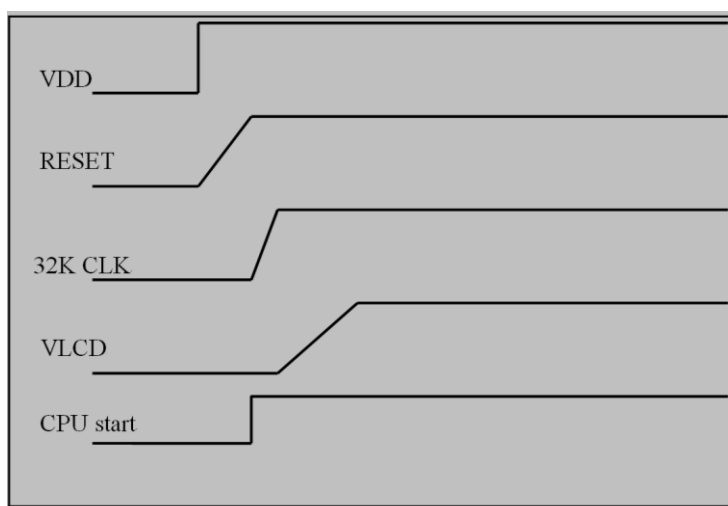
Power on timing sequence diagram



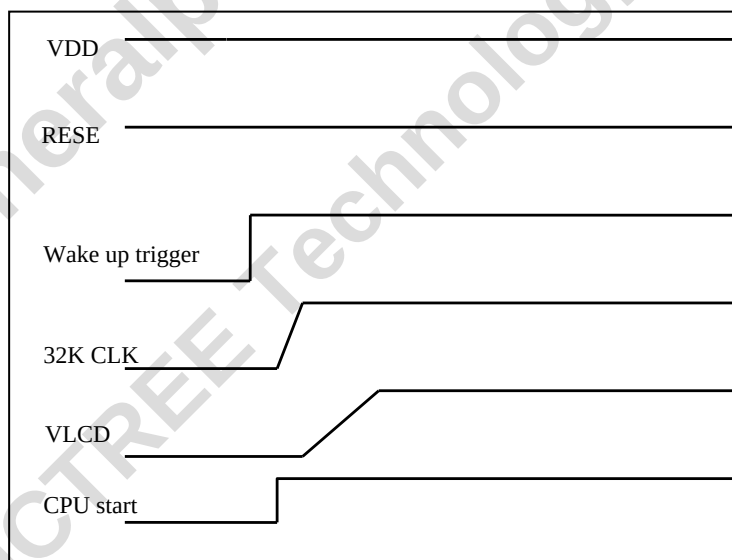
Wake up timing sequence diagram

GPL813P01X - Power ON/Stand-by Wakeup Crystal 32K Start-up Time + VLCD stable time			
Condition: 1/4 bias			
	X32KStart-up	VLCD stable time	Total time
VDD=4.5V	350ms	80ms	430ms
VDD=3.0V	130ms	80ms	210ms

Note: When powering on or waking up the CPU from standby to restart, it takes 210~430ms to waiting turn on the LCD when the crystal oscillator 32K is selected.



Power on timing sequence diagram



Wake up timing sequence diagram

3 I/O CONFIGURATION

3.1 Introduction

GPL813P01X has 36 general IO pins which are grouped into three parts: PortA[3:0], PortB[7:0], PortC[7:0], PortD[7:0] and PortE[7:0]. All of these IO pin can configure as input or output function. And some of pins share with special function such as buzzer output, RFC output...etc.

In GPL813P01X, the PortA[3:0] can be configured as PWMIO and shared with SEG[18:21].

PortB[7:0] can be selected as LCD common [0:7] signals through code option.

PortC[3:0] shared with SEG[10:13], PortC[5:4] can be configured as X32K[X32O,X32I], PortC[7:6] shared with SEG[8:9].

PortD[1:0] can be configured as buzzer output and shared with SEG[30:31], PortD.2 / PortD.6 can be configured as external1 / external2 input INT by software programmable. PortD[5:3] can be configured as RFC OUT and shard with SEG[27:29], PortD[7] can be configured as clock 32K output and shared with SEG[26].

PortE[7:0] can be enabled for key change wake-up function to wake CPU up from sleep mode and shared with SEG[14:25].

3.2 I/O Port Shared with Other Function List

GPL813P01X PortA[3:0]: SEG: Code option, others: programming option.

PortA								
Bit	7	6	5	4	3	2	1	0
Special	-	-	-	-	PA3	PA2	PA1	PA0
Function	-	-	-	-	SEG18	SEG19	SEG20	SEG21
	-	-	-	-	PWMIO3	PWMIO2	PWMIO1	PWMIO0

GPL813P01X PortB[7:0]: COM: Code option.

PortB								
Bit	7	6	5	4	3	2	1	0
Special	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
Function	COM0	COM1	COM2	COM3	COM4	COM5	COM6	COM7

GPL813P01X PortC[7:0]: SEG/X32K: Code option.

PortC								
Bit	7	6	5	4	3	2	1	0
Special	PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
Function	SEG8	SEG9	X32O	X32I	SEG10	SEG11	SEG12	SEG13

GPL813P01X PortD[7:0]: SEG: Code option, others: programming option.

PortD								
Bit	7	6	5	4	3	2	1	0
Special	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0
Function	SEG26	-	SEG27	SEG28	SEG29	-	SEG30	SEG31
	Clock 32K out	EXT2_INT	RFC OUT	RFC OUT	RFC OUT	EXT1_INT RFC IN	Buzzer	Buzzer

GPL813P01X PortE[7:0] : SEG: Code option.

PortE								
Bit	7	6	5	4	3	2	1	0
Special	PE7	PE6	PE5	PE4	PE3	PE2	PE1	PE0
Function	SEG14	SEG15	SEG16	SEG17	SEG22	SEG23	SEG24	SEG25
	Key wake-up	Key Wake-up	Key Wake-up	Key Wake-up	Key Wake-up	Key Wake-up	Key Wake-up	Key Wake-up

3.3 PortA Control Register Description

PortA can be several modes via I/O control and direct register configuration; please refer to the following table for more information about the relationship between register configuration and I/O mode.

Attribution (0x0006)	Direction (0x0000)	Data (0x0002)	Function	
0	0	0	Input with pull-low	
0	0	1	Input mode, float state	
0	1	0	Output low	PWM
0	1	1	Output high	PWM
1	1	0	Pad floating	

3.4 \$0000, P_IO_PortA_Dir

P_IO_PortA_Dir		0x0000				IOA Direction Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	IOADIR			
Default	-	-	-	-	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOADIR	W	This control register sets the direction of I/O PortA.	0: input 1: output

- Input with pull low mode: EV 813V00A:160K-Ohm pull down resistors
- Input with pull low mode: Real 813P01X: 55K-Ohm pull down resistors
- Output mode: pure buffer

3.5 \$0002, P_IO_PortA_Data

P_IO_PortA_Data			0x0002			IOA Data Register		
Bit	7	6	5	4	3	2	1	0
Function	-				IOADATA			
Default	-	-	-	-	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOADATA	R/W	This control register sets the data of I/O PortA.	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: not change IO status

3.6 PortB Control Register Description

PortB can be several modes via I/O control and direct register configuration; please refer to the following table to clear relationship between register configuration and I/O mode.

Attribution (0x0014)	Direction (0x0010)	Data (0x0012)	Function
0	0	0	Input with pull-low
0	0	1	Input mode, float state
0	1	0	Output low
0	1	1	Output high
1	1	0	Pad floating

3.7 \$0010, P_IO_PortB_Dir

P_IO_PortB_Dir			0x0010			IOB Direction Register		
Bit	7	6	5	4	3	2	1	0
Function	IOBDIR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOBDIR	W	This control register sets the direction of I/O PortB.	0: input 1: output

- Input with pull low mode: EV 813V00A:160KΩ pull down resistors
- Input with pull low mode: Real 813P01X: 55K-Ohm pull down resistors
- Output mode: pure buffer

3.8 \$0014, P_IO_PortB_Attrib

P_IO_PortB_Attrib			0x0014				IOB Attribution Register	
Bit	7	6	5	4	3	2	1	0
Function	IOBAttrib							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOBAttrib	W	This control register sets the attribution of I/O PortB.	1: Special function I/O 0: General I/O

3.9 \$0012, P_IO_PortB_Data

P_IO_PortB_Data			0x0012				IOB Data Register	
Bit	7	6	5	4	3	2	1	0
Function	IOBDATA							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOBDATA	R/W	This control register sets the data of I/O PortB.	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: not change IO status

3.10 PortC Control Register Description

The PortC can be several modes via I/O control and direct register configuration; user can refer to the following table for more information about register configuration and I/O mode.

Attribution (0x0006)	Direction (0x0001)	Data (0x0005)	Function
0	0	0	Input with pull-low
0	0	1	Input mode, floating state
0	1	0	Output low
0	1	1	Output high
1	1	0	Pad floating

3.11 0x0001, P_IO_PortC_Dir

P_IO_PortC_Dir			0x0001				IOC Direction Register	
Bit	7	6	5	4	3	2	1	0
Function	IOCDIR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOCDIR	R/W	This control register sets the direction of I/O PortC.	0: input 1: output

- Input with pull low mode: EV 813V00A:160KΩ pull down resistors
- Input with pull low mode: Real 813P01X: 55K-Ohm pull down resistors
- Output mode: pure buffer

3.12 0x0005, P_IO_PortC_Data

P_IO_PortC_Data			0x0005			IOC Data Register		
Bit	7	6	5	4	3	2	1	0
Function	IOCDATA							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOCDATA	R/W	This control register sets the data of I/O PortC.	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: IO status not changed

3.13 PortE Control Register Description

PortE can be several modes via I/O control and direct register configuration; user can refer to the following table for more information about the relationship between register configuration and I/O modes.

Attribution (0x0006)	Direction (0x000B)	Data (0x0003)	Function
0	0	0	Input with pull-low
0	0	1	Input mode, floating state
0	1	0	Output low
0	1	1	Output high
1	1	0	Pad floating

3.14 0x000B, P_IO_PortE_Dir

P_IO_PortE_Dir			0x000B			IOE Direction Register		
Bit	7	6	5	4	3	2	1	0
Function	IOEDIR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOEDIR	R/W	This control register sets the direction of I/O PortE.	0: input 1: output

3.15 0x0003, P_IO_PortE_Data

P_IO_PortE_Data			0x0003			IOE Data Register		
Bit	7	6	5	4	3	2	1	0
Function	IOEDATA							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOEDATA	R/W	This control register sets the data of I/O PortE.	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: IO status not changed

3.16 0x0004, P_IO_PortE_DataLatch

P_IO_PortE_DataLatch			0x0004			IOE Pin Latch Register		
Bit	7	6	5	4	3	2	1	0
Function	IOELATCH							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	IOELATCH	R	Data latch for key change wakeup. Reading this port latches the data of IOE [7:0] status.	IOE supports key change wake-up function. If key change wake-up function is used, read this register before CPU enters into standby or halt mode.

3.17 0x0006, P_IO_Ctrl

P_IO_Ctrl			0x0006			IO Attribute Control Register		
Bit	7	6	5	4	3	2	1	0
Function	PA[3:2]	PA[1:0]	PC[7:4]	PC[3:0]	PE[7:4]	PE[3:2]	PE1	PE0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	PA[3:2]	W	Attribute control register	0: General I/O 1: Special function I/O
6	PA[1:0]	W	Attribute control register	0: General I/O

Bit	Function	Type	Description	Condition
				1: Special function I/O
5	PC[7:4]	W	Attribute control register	0: General I/O 1: Special function I/O
4	PC[3:0]	W	Attribute control register	0: General I/O 1: Special function I/O
3	PE[7:4]	W	Attribute control register	0: General I/O 1: Special function I/O
2	PE[3:2]	W	Attribute control register	0: General I/O 1: Special function I/O
1	PE1	W	Attribute control register	0: General I/O 1: Special function I/O
0	PE0	W	Attribute control register	0: General I/O 1: Special function I/O

Note1: When using key matrix, scan key must be output high (sensing) or input floating (non-sensing). If scan key is set as output high (sensing) and output low (non-sensing), I/O will conflict and cause more power consumption when pressing two keys at the same time.

3.18 PortD Control Register Description

The PortD can be several modes via I/O control and direct register configuration. Refer to the following table to clear relationship between register configuration and I/O mode.

Attribution (0x0015)	Direction (0x0011)	Data (0x0013)	Function	
0	0	0	Input with pull-low	EXT1, EXT2
0	0	1	Input floating state	RFC Input (PD2)
0	1	0	Output low	C32KOUT, Buzzer RFC Output PD[5:3]
0	1	1	Output high	C32KOUT, Buzzer
1	1	0	Pad floating	RFC Output PD[5:3]

3.19 0x0011, P_IO_PortD_Dir

P_IO_PortD_Dir			0x0011			IOD Direction Register		
Bit	7	6	5	4	3	2	1	0
Function	IODDIR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IODDIR	W	This control register sets the direction of I/O PortD.	0: input 1: output

3.20 0x0015, P_IO_PortD_Attrib

P_IO_PortD_Attrib			0x0015			IOD Direction Register		
Bit	7	6	5	4	3	2	1	0
Function	IODAttrib							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IODAttrib	W	Attribute control register	1: Special function I/O 0: General I/O

Note: In RFC application, PD3/PD4/PD5 will be set as 1 (pad-floating).

3.21 0x0013, P_IO_PortD_Data

P_IO_PortD_Data			0x0013			IOD Data Register		
Bit	7	6	5	4	3	2	1	0
Function	IODDATA							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IODDATA	R/W	This control register sets the data of I/O PortD.	Read: read pin current status Write: IO status not changed

3.22 0x0016, P_AUDIO_BZ_Ctrl

P_AUDIO_BZ_Ctrl			0x0016			IOD Buzzer Control Register		
Bit	7	6	5	4	3	2	1	0
Function	C32KOUT	-					Buzzer1	Buzzer0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	C32KOUT	W	This bit sets PortD.b7 as normal I/O or output 32KHz clock.	0 : PD7 as normal IO 1 : PD7 output CLK32K
[6:2]	Reserved	-	-	-
[1:0]	Buzzer	W	This control register sets PortD[1:0] as buzzer I/O or normal I/O.	11: PortD[1:0] as buzzer output 00: PortD[1:0] as normal output PortD[1:0] must be set output low / high first, and then set buzzer output. User can set PortD[1] as buzzer output, PortD[0] as I/O and vice versa.

3.23 I/O Structure

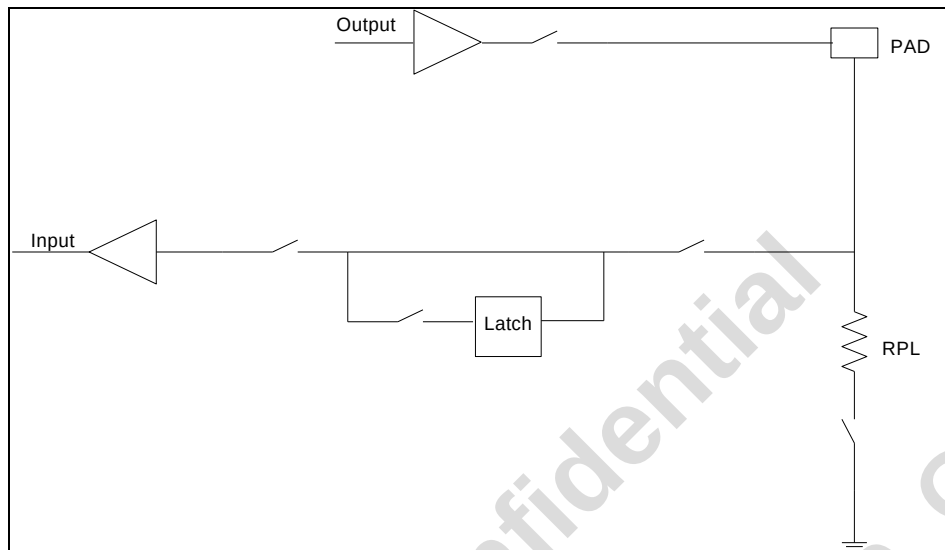


Fig2.1 IO PC/PD/PE Structure

4 MEMORY MAPPING

4.1 Introduction

GPL813P01X includes 256 bytes SRAM, 32 bytes LCD DPRAM and 8K bytes OTP ROM sizes based on various ICs.

4.2 GPL813P01X Memory Mapping

GPL813P01X MEMORY MAPPING : 8KB OTPROM, 256B SRAM.

CPU view	
\$0000~\$007F	IO & Reg
\$0080~\$00FF	RAM
\$0100~\$017F	Reserved
\$0180~\$01FF	Stack RAM
\$0200~\$3DFF	Reserved
\$3E00~\$3E73	LCD DPRAM
\$3E74~\$DFFF	Reserved
\$E000~\$FFDF	OTP ROM
\$FFE0~\$FFF9	GP used reserved
\$FFFA~\$FFFF	Interrupt Vector

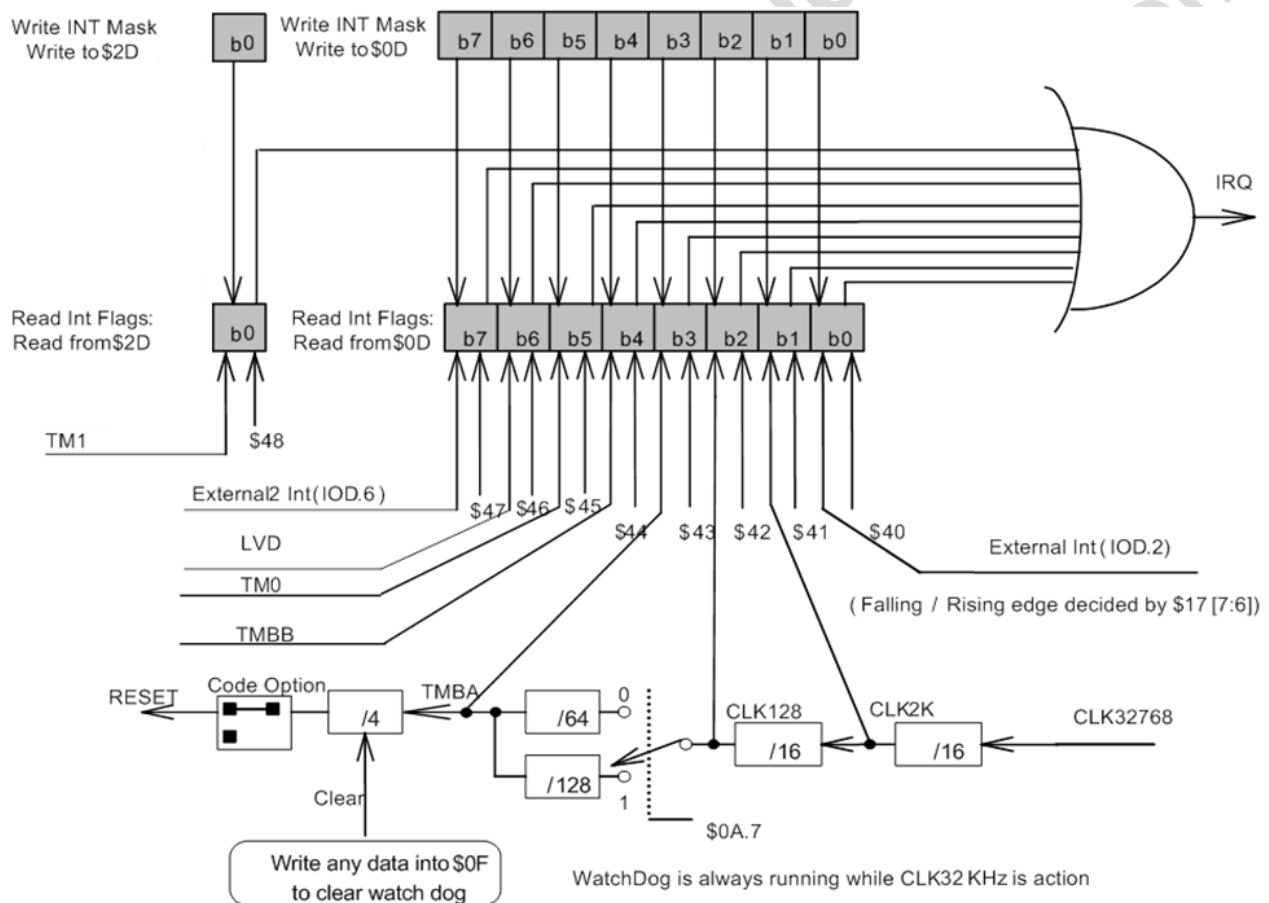
In GPL813P01X,

- Stack Area: 0x180~0x1FF is for CPU stack area, Programmers can use TXS instruction to assign Stack option start address.
- User's program interrupt vector: 0xFFFA ~ 0xFFFF
- Area reserved for Generalplus used: 0xFFE0 ~ 0xFFF9

5 INTERRUPT / WATCHDOG CONFIGURATION

5.1 Introduction

GPL813P01X supports 9 interrupt sources: CLK2KHz, CLK128Hz, TMBA, TMBB, EXT1, EXT2, LVD, TM0, and TM1. All interrupts can be ON/OFF separately through setting 0x000D and 0x002D control register if user utilizes timer count as interrupt source, such as CLK2KHz, CLK128Hz, TimeBaseA and TimeBaseB. These timer count clock sources are coming from CLK32768, but CLK32768 is not equal to 32768Hz. The CLK32768 clock source can be selected from ROSC 32768Hz $\pm 20\%$ or external 32768Hz crystal OSC. It depends on CLK32768 clock source code option. GPL813P01A also has a built-in watchdog function which can be enabled by code option. If watchdog function is code option enabled and it is not cleared within 1.5 seconds, a system reset event will occur. We suggest user clear watchdog within one second.



TMBB: 4 Hz, 8 Hz, 16 Hz, 32 Hz, 64 Hz, 128 Hz, 1024 Hz, 4096 Hz depend on \$0A b0, b1 and b2

\$0A(W)	b2	b1	b0	
0	0	0	0	4 Hz Interrupt
0	0	1	0	8 Hz Interrupt
0	0	0	1	16 Hz Interrupt
0	1	1	1	32 Hz Interrupt
1	0	0	0	64 Hz Interrupt
1	1	1	0	128 Hz Interrupt
1	0	1	1	1024 Hz Interrupt
1	1	1	1	4096 Hz Interrupt

Note:

- 1: Changing \$000A not only changes interrupt time, but also wake-up time.
- 2: Do not forget to clear watchdog counter if watchdog is activated while 32KHz running.
- 3: Writing any data to \$0040 clears interrupt bit0 (EXT1).
- 4: Writing any data to \$0041 clears interrupt bit1 (CLK2KHz).
- 5: Writing any data to \$0042 clears interrupt bit2 (CLK128Hz).
- 6: Writing any data to \$0043 clears interrupt bit3 (TMBA).
- 7: Writing any data to \$0044 clears interrupt bit4 (TMBB).
- 8: Writing any data to \$0045 clears interrupt bit5 (TM0).
- 9: Writing any data to \$0046 clears interrupt bit6 (LVD).
- 10: Writing any data to \$0047 clears interrupt bit7 (EXT2).
- 11: Writing any data to \$0048 clears interrupt2 bit0 (TM1).

5.2 0x000D, P_INT_Ctrl

P_INT_Ctrl			0x000D			Interrupt Control Register		
Bit	7	6	5	4	3	2	1	0
Function	EXT2_INT	LVD_INT	TM0_INT	TMBB_INT	TMBA_INT	128Hz_INT	2KHz_INT	EXT1_INT
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	EXT2_INT (IOD.6)	R/W	Writing this bit can enable/disable external2 interrupt. Reading this bit can appear external2 interrupt status. Writing any data to 0x0047 can clear this interrupt flag.	Read 0: external2 interrupt cleared Read 1: external2 interrupt occurred Write 0: disable external2 interrupt Write 1: enable external2 interrupt
6	LVD_INT	R/W	Writing this bit can enable/disable LVD INT. Reading this bit can appear LVD INT status Writing any data to 0x0046 can clear this interrupt flag.	Read 0: LVD INT cleared Read 1: LVD INT occurred Write 0: LVD INT disabled Write 1: LVD INT disabled
5	TM0_INT	R/W	Writing this bit can enable/disable timer0 overflow INT. Reading this bit can appear timer0 overflow INT status. About timer configuration, please refer to <u>Timer Chapter</u> . Writing any data to 0x0045 can clear this interrupt flag.	Read 0: timer0 overflow INT cleared Read 1: timer0 overflow INT occurred Write 0: timer0 overflow INT disabled Write 1: enable timer0 overflow INT
4	TMBB_INT	R/W	Writing this bit can enable/disable TimeBaseB INT. Reading this bit can appear TimeBaseB	Read 0: TimeBaseB INT cleared Read 1: TimeBaseB INT occurred Write 0: TimeBaseB INT disabled

Bit	Function	Type	Description	Condition
			INT status The TimeBaseB count frequency sets by 0x000A.[2:0]. Writing any data to 0x0044 can clear this interrupt flag.	Write 1: TimeBaseB INT enabled
3	TMBA_INT	R/W	Writing this bit can enable/disable TimeBaseA count INT. Reading this bit to obtain TimeBaseA INT status. User can set TimeBaseA as 1 or 2Hz interrupt by 0x000A.b7. Writing any data to 0x0043 can clear this interrupt flag.	Read 0: TimeBaseA INT cleared Read 1: TimeBaseA INT occurred Write 0: TimeBaseA INT disabled Write 1: TimeBaseA INT enabled
2	128Hz_INT	R/W	Writing this bit can enable/disable 128Hz count INT. Reading this bit can appear 128Hz count INT status. Writing any data to 0x0042 can clear this interrupt flag.	Read 0: 128Hz count INT cleared Read 1: 128Hz count INT occurred Write 0: 128Hz INT disabled Write 1: 128Hz INT enabled
1	2KHz_INT	R/W	Writing this bit can enable/disable 2048Hz count INT. Reading this bit can appear 2048Hz count INT status. Writing any data to 0x0041 can clear this interrupt flag.	Read 0: 2048Hz count INT cleared Read 1: 2048Hz count INT occurred Write 0: 2048Hz INT disabled Write 1: 2048Hz INT enabled
0	EXT1_INT (IOD.2)	R/W	Writing this bit can enable/disable external1 interrupt. Reading this bit can appear external1 interrupt status. Writing any data to 0x0040 can clear this interrupt flag.	Read 0: external1 interrupt cleared Read 1: external1 interrupt occurred Write 0: external1 interrupt disabled Write 1: external1 interrupt enabled

5.3 \$002D, P_INT_Ctrl2

P_INT_Ctrl2			0x002D			Interrupt Control2 Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	-	-	-	TM1_INT
Default	-	-	-	-	-	-	-	0

Bit	Function	Type	Description	Condition
[7:1]	-	-	Reserved	-
0	TM1	R/W	Writing this bit can enable/disable timer1 overflow INT.	Read 0: timer1 overflow INT cleared

Bit	Function	Type	Description	Condition
			Reading this bit can appear timer1 overflow INT status. About timer configuration, please refer to Timer Chapter . Writing any data to 0x0048 can clear this interrupt flag.	Read 1: timer1 overflow INT occurred Write 0: timer1 overflow INT disabled Write 1: timer1 overflow INT enabled

5.4 \$000A, P_TIMER_TimeBase_Sel

The P_TIMER_TimeBase_Sel(0x000A) supports two functions, one is to set interrupt count source through writing this register, and the other is to make time counter through reading this register.

P_TIMER_TimeBase_Sel		0x000A			TimeBase Clock Selection Register			
Bit	7	6	5	4	3	2	1	0
Function	TMBA R0.5Hz	R1Hz	R2Hz	R4Hz	R8Hz	TMBBSEL R16Hz	TMBBSEL R32Hz	TMBBSEL R64Hz
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	TMBA	W	This bit is used to select TimeBaseA 1 or 2Hz as interrupt source. The corresponding interrupt control register is 0x000D.b3.	Write 0: selects TimeBaseA 2Hz Write 1: selects TimeBaseA 1Hz
7	R0.5Hz	R	After 0.5Hz timer count is done, this bit will be set as 1 and keep until 1/64 seconds. Then, it will be cleared by hardware automatically.	Reading this register will acquire timer count value. The each step unit is 64Hz. The register count is from 0x00 → 0x01 → 0x02 ... 0xFF → 0x00 repetitiously.
6	R1Hz	R	After 1Hz timer count is done, this bit will be set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
5	R2Hz	R	After 2Hz timer count is done, this bit will be set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
4	R4Hz	R	After 4Hz timer count is done, this bit will be set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
3	R8Hz	R	After 8Hz timer count is done, this bit will be set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
2	R16Hz	R	After 16Hz timer count is done, this bit will be set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
1	R32Hz	R	After 32Hz timer count is done, this bit will be set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
0	R64Hz	R	After 64Hz timer count is done, this bit will be	

Bit	Function	Type	Description	Condition
			set as 1 and kept for 1/64 seconds. Then, it will be cleared by hardware automatically.	
[2:0]	TMBBSEL	W	These bits are used to select TimeBaseB frequency as the interrupt source. The corresponding interrupt control register is 0x000D.b4. In real chip(GPL813P01X) not support TimerBase Wake up "111": 4096Hz	TimeBaseB frequency selection Write 000: 4Hz Write 001: 8Hz Write 010: 16Hz Write 011: 32Hz Write 100: 64Hz Write 101: 128Hz Write 110: 1024Hz Write 111: 4096Hz

Note:

1. P_TIMER_TimeBase_Sel \$000A not only changes the interrupt time, but also wake-up time.
2. Do not forget to clear watchdog counter if watchdog is activated while CLK32768 clock running.
- (a). Suppose \$000A is read as "00" initially. After a certain period of time, \$000A is read as "40". Therefore, 1.0 second is the time between "00" and "40".
- (b). Suppose \$000A is read as "01" at the beginning. Then, \$000A is read as "05".

We know, $\frac{(((1/64)+(1/16)) - (1/64))}{\text{second}} = 1/16\text{seconds}$.

$\uparrow$ $\uparrow$
 05 01

3. CLK32768 clock source is from ROSC 32768Hz $\pm 20\%$ or external 32768Hz crystal.

5.5 \$000E, P_Reset_Status

P_Reset_Status			0x000E			Reset Status Register		
Bit	7	6	5	4	3	2	1	0
Function	-						FlagON	FlagC
Default	-	-	-	-	-	-	0	0

Bit	Function	Type	Description	Condition
[7:2]	-	-	Reserved	-
1	FlagON	R/W	When in Standby mode and Reset pin is tied to low, this status bit will be set.	Read 0: Reset status no change. Read 1: Reset status occurs from sleep mode. Write 0: clear sleep mode status. Write 1: NC
0	FlagC	R/W	When in normal mode or halt mode and Reset pin is tied to low, this status bit will be set.	Read 0: Reset status no change. Read 1: Reset status occurs from normal mode or halt mode. Write 0: clear normal mode or halt mode status. Write 1: NC

Note: When Reset pin is tied to low, it will generate a system reset. The reset does not reset LCD control register configuration and **P_Reset_Status** will be set.

5.6 \$000F, P_WDT_Clear

P_WDT_Clear			0x000F			Watchdog Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	WDTCLR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	WDTCLR	W	Writing any data to this register will clear watchdog count.	-

Note: GENERALPLUS highly recommends clearing WDT every 1 second to avoid accidental reset event.

5.7 \$0040, P_INT_Ext1_Clear

P_INT_Ext1_Clear			0x0040			External1 Interrupt Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	EXT1INT_Clr							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	EXT1INT_Clr	W	Writing any data to this register will clear external1 interrupt flag.	-

5.8 \$0041, P_INT_2KHz_Clear

P_INT_2KHz_Clear			0x0041			2KHz Interrupt Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	CLK2K_Clr							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	CLK2K_Clr	W	Writing any data to this register will clear 2KHz interrupt flag.	-

5.9 \$0042, P_INT_128Hz_Clear

P_INT_128Hz_Clear			0x0042			128Hz Interrupt Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	CLK128_Clr							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	CLK128_Clr	W	Writing any data to this register will clear 128Hz interrupt flag.	-

5.10 \$0043, P_INT_TimeBaseA_Clear

P_INT_TimeBaseA_Clear			0x0043			TimeBaseA Interrupt Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	TMBACLr							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TMBACLr	W	Writing any data to this register will clear TimeBaseA interrupt flag.	-

5.11 \$0044, P_INT_TimeBaseB_Clear

P_INT_TimeBaseB_Clear			0x0044			TimeBaseB Interrupt Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	TMBBCLr							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TMBCLR	W	Writing any data to this register will clear TimeBaseB interrupt flag.	-

5.12 \$0045, P_INT_Timer0_Clear

P_INT_Timer0_Clear			0x0045		Timer0 Overflow Interrupt Flag Clear Register			
Bit	7	6	5	4	3	2	1	0
Function	TM0CLR							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM0CLR	W	Writing any data to this register will clear timer0 overflow interrupt flag.	-

5.13 \$0046, P_INT_LVD_Clear

P_INT_LVD_Clear			0x0046		LVD Interrupt Flag Clear Register			
Bit	7	6	5	4	3	2	1	0
Function	LVDCLR							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	LVDCLR	W	Writing any data to this register will clear LVD interrupt flag.	-

5.14 \$0047, P_INT_Ext2_Clear

P_INT_Ext2_Clear			0x0047		External2 Interrupt Flag Clear Register			
Bit	7	6	5	4	3	2	1	0
Function	EXT2INT_Clr							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	EXT2INT_Clr	W	Writing any data to this register will clear external2 interrupt flag.	-

5.15 \$0048, P_INT_Timer1_Clear

P_INT_Timer1_Clear			0x0048		Timer1 Overflow Interrupt Flag Clear Register			
Bit	7	6	5	4	3	2	1	0
Function	TM1CLR							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM1CLR	W	Writing any data to this register will clear timer1 overflow interrupt flag.	-

5.16 Interrupt Programming Example

(a). Clear all interrupts

```
LDA    #%00000000    ;
STA    P_INT_Ctrl    ; Clear Port$0D (Interrupt Port)
STA    IntFlags      ; Store into a RAM
```

(b). Enable interrupts

```
LDA    #%00001000    ; Enable 2Hz Int.
STA    P_INT_Ctrl
STA    IntFlags      ; Store into a RAM
```

(c). Interrupt Process:

IRQ:

```
LDA    $0D
STA    R_Temp0D      ; Store into a temporary RAM for other interrupt checks
AND    #%00001000
BEQ    L_Check128Int
```

L_T2HzInt:

```
STA    P_INT_TimeBaseA_Clear
....
/* Process T2Hz Function */
```

L_Chk128Int:

```
LDA    R_Temp0D
AND    #%00000100
BEQ    L_ChkCLK2K
```

L_T128HzInt:

```
STA    P_INT_128Hz_Clear
....
/* Process T128Hz Function */
```

L_ChkCLK2K:

```
AND    #%00000010
BEQ    L_ExitIRQ
```

L_CLK2K:

```
STA    P_INT_2KHz_Clear
```



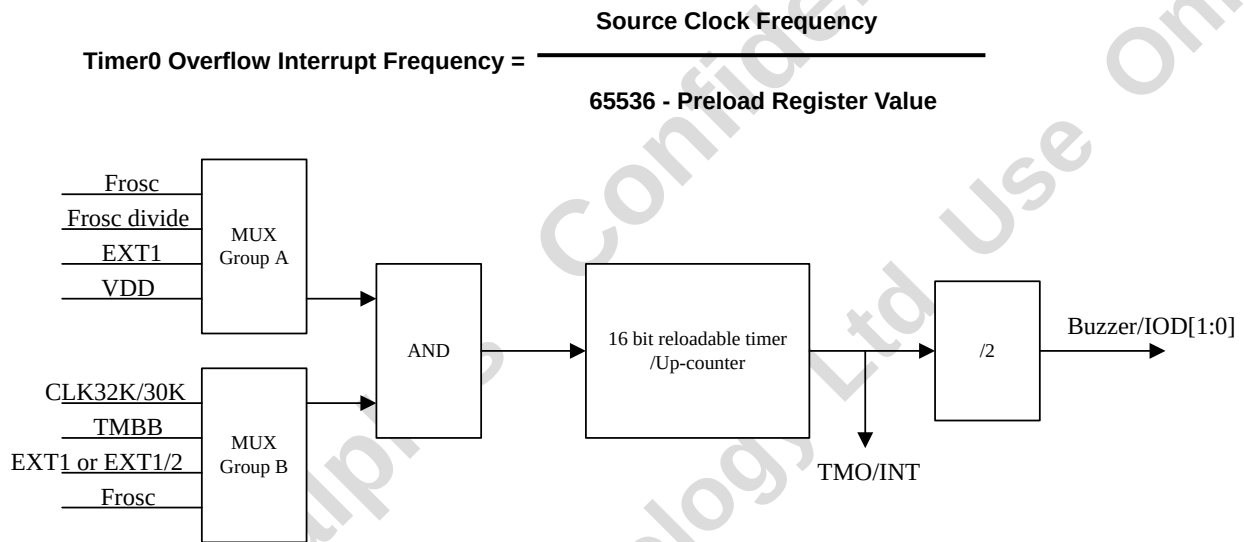
```
....  
/* Process CLK2K Function */  
L_ExitIRQ:  
RTI
```

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6 TIMER

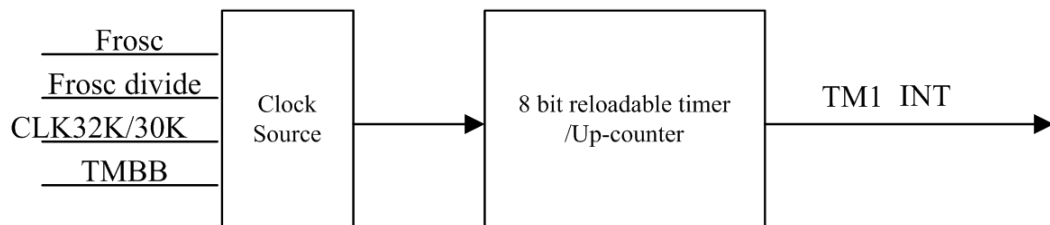
6.1 Introduction

GPL813P01X includes 16-bit timer and 8-bit up-count timers. The clock source of the timer comes from internal Frosc, Frosc division, CLK32768, TimerbaseB, and external1 clock (inputted from PD2). The 16-bit timer has an up-counter and it increases on the rising or falling edge of internal clock source. Timer supports external1 clock input. Initial value of the up-counter is stored in the preloaded register. When the timer is enabled, overflow occurs and writing any value into P_TIMER_TM0Load_Ctrl register will reload into timer counter. For example, if the preloaded register value is 0xFFFFC, the timer will count in the sequence order of 0xFFFFC, 0xFFFFD, 0xFFFFE, 0xFFFFF, overflow and then reload timer counter 0xFFFFC, 0xFFFFD, and etc.



Source Clock Frequency

Timer1 Overflow Interrupt Frequency = $\frac{\text{Source Clock Frequency}}{256 - \text{Preload Register Value}}$



6.2 \$0022, P_TIMER_Timer0_Ctrl

P_TIMER_Timer0_Ctrl			0x0022		Timer0 Control Register			
Bit	7	6	5	4	3	2	1	0
Function	TMR0EN	-	EXTA	-	TCKB1	TCKB0	TCKA1	TCKA0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	TMR0EN	R/W	This bit sets timer0 as enable/disable.	0: disabled 1: enabled
6	-	-	This bit must fix to "1"	Fixed to "1"
5	EXTA	R/W	This bit sets group B clock source EXTA as EXT1 or EXT1/2 as timer clock. The EXT1 clock input is from PD.2 pin.	0: EXTA=EXT1 as timer clock, 1: EXTA=EXT1/2 as timer clock;
4	-		Reserved	-
[3:2]	TCKB	R/W	These bits select group B as timer0 clock source. The TimebaseB clock selection, please see 0x000A[2:0] register configuration. If use group B as timer clock source, please set group A as VDD.	00→CLK32KHz / ROOSC32KHz 01→TMBB (0x000A[2:0]) 10→EXTA 11→VDD
[1:0]	TCKA	R/W	These bits select group A as timer0 clock source. The system clock and system clock divide selection. Please refer to CPU CLOCK SELECTIONS. If use group A as timer clock source, please set group B as VDD.	00→Frosc (system clock) 01→Frosc divide 10→EXT1 (IOD2) 11→VDD

6.3 \$0020, P_TIMER_TM0Data_LB

P_TIMER_TM0Data_LB		0x0020				Timer0 Count Low Byte Data Register		
Bit	7	6	5	4	3	2	1	0
Function	TM0LB_Data							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM0LB_Data	R/W	Timer0 count low byte value configuration. It is an up-count timer. When count is overflow, the count value will reload automatically.	Write: set timer0 count value to buffer Read: read current up-count value

6.4 \$0021, P_TIMER_TM0Data_HB

P_TIMER_TM0Data_HB		0x0021				Timer0 Count High Byte Data Register		
Bit	7	6	5	4	3	2	1	0
Function	TM0HB_Data							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM0HB_Data	R/W	Timer0 count high byte value configuration. It is an up-count timer. When count is overflow, the count value will reload automatically.	Write: set timer0 count value to buffer Read: read current up-count value

6.5 \$0023, P_TIMER_TM0Load_Ctrl

P_TIMER_TM0Load_Ctrl			0x0023			Timer0 Count Reload Register		
Bit	7	6	5	4	3	2	1	0
Function	TM0RELOAD							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM0RELOAD	W	Timer0 count reload control register.	Writing any data to this register will make up-count value reloaded.

6.6 \$0026, P_TIMER_Timer1_Ctrl

P_TIMER_Timer1_Ctrl			0x0026			Timer1 Control Register		
Bit	7	6	5	4	3	2	1	0
Function	TMR1EN	-	-	-	-	-	TM1CK1	TM1CK0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	TMR1EN	R/W	This bit sets timer1 as enable/disable.	0: disabled 1: enabled
[6:2]	-	-	Reserved	-
[1:0]	TM1CK	R/W	These bits select timer1's clock source. For TimebaseB clock selection, please see 0x000A[2:0] register configuration. The system clock and system clock divide selection. Please refer to CPU CLOCK SELECTIONS.	00→CLK32KHz / ROsc32KHz 01→TMBB (0x000A[2:0]) 10→Frosc (system clock) 11→Frosc division

6.7 \$0025, P_TIMER_TM1Data

P_TIMER_TM1Data			0x0025			Timer1 Count Data Register		
Bit	7	6	5	4	3	2	1	0
Function	TM1 Data							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM1_Data	R/W	Timer1 count value configuration It is an up-count timer. When count is overflowed, the count value will reload automatically.	Write: set timer1 count value to buffer Read: read current up-count value

6.8 \$0027, P_TIMER_TM1Load_Ctrl

P_TIMER_TM1Load_Ctrl			0x0027			Timer1 Count Reload Register		
Bit	7	6	5	4	3	2	1	0
Function	TM1RELOAD							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	TM1RELOAD	W	Timer1 count reload control register	Writing any data to this register will make up-count value reloaded.

7 CPU CLOCK SELECTIONS

7.1 Introduction

GPL813P01X system clock only 4000KHz. GPL813V00A system clock has the following programmable options: 225KHz, 500KHz, 1000KHz, 1800KHz, and 4000KHz.

7.2 \$0018, P_CLK_CPU_Ctrl

P_CLK_CPU_Ctrl			0x0018		CPU Clock Register			
Bit	7	6	5	4	3	2	1	0
Function	SYSCLK2	SYSCLK1	SYSCLK0	-	-	CPUCK2	CPUCK1	CPUCK0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:5]	SYSCLK	R/W	This bit can set system frequency, Frosc= System clock	0 0 0: Frosc=225KHz 0 1 0: Frosc=500KHz 1 0 0: Frosc=1800KHz 1 1 0: Frosc=1000KHz X X 1: Frosc=4000KHz GPL813V00A used only. X X X: Frosc=4000KHz GPL813P01X used only.
[4:3]	-	-	Reserved	-
[2:0]	CPUCK	R/W	This bit can set CPU frequency F _{CPU} = CPU clock frequency Frosc= system frequency	0 0 0: F _{CPU} =Frosc 0 0 1: F _{CPU} =Frosc / 2 0 1 0: F _{CPU} =Frosc / 4 0 1 1: F _{CPU} =Frosc / 8 1 x x: F _{CPU} =Frosc / 16

Note: If user wants **EV chip GPL813V00A** to set high speed frequency **over 500KHz**, the setting sequence refer to following steps.

1. Set Fcpu = (225K or 500KHz) / 16
2. Set Fcpu = (4MHz) / 16
3. Set Fcpu = (4MHz) / 1

Ex:

```
LDA    #x1xx1xxb
STA    P_CLK_CPU_Ctrl    ; Set Fcpu = (500KHz) / 16
LDA    #xx1xx1xxb
STA    P_CLK_CPU_Ctrl    ; Set Fcpu = (4MHz) / 16
LDA    # xx1xx000b
STA    P_CLK_CPU_Ctrl    ; Set Fcpu = (4MHz) / 1
```

7.3 \$001A, P_CLK_HFDIV_Ctrl

P_CLK_HFDIV_Ctrl			0x001A		System Clock Divider Register			
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	-	-	DIV1	DIV0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:2]	-	-	Reserved	-
[1:0]	DIV	R/W	This frequency offers timer count.	0 0: Frosc_div = Frosc / 2 0 1: Frosc_div = Frosc / 4 1 0: Frosc_div = Frosc / 8 1 1: Frosc_div = Frosc / 16

8 LCD CONFIGURATION

8.1 Introduction

The DPRAM \$3E00 - \$3E73 are reserved for LCD display buffer. Simply write the corresponding bit in display buffer to display a pattern on LCD. The following table is the mapping of LCD buffer RAM. The DPRAM can be read and written. There is no list in following table address such as \$3E04-\$3E0F, \$3E14-\$3E1F, \$3E24-\$3E2F, \$3E34-\$3E3F, \$3E44-\$3E4F, \$3E54-\$3E5F, \$3E64-\$3E6F and \$3E74-\$3E7F; these addresses are invalid. User is not allowed using these areas.

GPL813P01X LCD Table:

	SEG0 - SEG7	SEG8 - SEG15	SEG16 - SEG23	SEG24 - SEG31
	b0-b7	b0-b7	b0-b7	b0-b7
Com0	\$3E00	\$3E01	\$3E02	\$3E03
Com1	\$3E10	\$3E11	\$3E12	\$3E13
Com2	\$3E20	\$3E21	\$3E22	\$3E23
Com3	\$3E30	\$3E31	\$3E32	\$3E33
Com4	\$3E40	\$3E41	\$3E42	\$3E43
Com5	\$3E50	\$3E51	\$3E52	\$3E53
Com6	\$3E60	\$3E61	\$3E62	\$3E63
Com7	\$3E70	\$3E71	\$3E72	\$3E73

Note:

1. If register's bits are not defined, they will not be implemented in the real chip and the readout value should be random.
2. ACON reset will not affect LCD control register (\$07) and all LCD RAM value (\$3E00~\$3E73) to keep LCD display quality.
3. GENERALPLUS recommends the speed of updating segment data shall be slower than 200Hz.

8.2 \$0007, P_LCD_Ctrl

P_LCD_Ctrl			0x0007		LCD Control Register			
Bit	7	6	5	4	3	2	1	0
Function	Duty1	Duty0	BIAS0	BIAS1	Duty2	Duty3	-	LCD EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6, 3:2]	DUTY	W	LCD duty setup	Duty[3:0]: 0110: 1/8 duty 0111: 1/6 duty 1000: 1/5 duty 1001: 1/4 duty 1010: 1/3 duty
[5:4]	BIAS	W	This bit sets LCD bias voltage.	BIAS[1:0] 00: 1/4 bias 01: 1/3 bias 10: 1/2 bias 11: Reserved
1	-	-	Reserved	-
0	LCD EN	W	This bit enables or disables LCD.	0: LCD display disabled 1: LCD display enabled

Note: In GPL813P01X, the COM[0:7] can be selected as PB[7:0] through code option. (PB[7:0] code option as COM[0:7] and register 0x0007 set as 1/8, 1/6, 1/5, 1/4, 1/3 duty only)

8.3 \$0019, P_LCD_VLCD_Ctrl

P_LCD_VLCD_Ctrl			0x0019		LCD Control Register			
Bit	7	6	5	4	3	2	1	0
Function	-		GREEN1	GREEN0	-		VLCD1	VLCD0
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[7:6]	-	-	Reserved	-
[5:4]	Halt Green		Must Write “10” or “11” to enter halt green mode, which consumes less halt current.	Halt green[5:4] 00: Reserved 01: Reserved 10: Halt current min 11: Halt current Normal
[3:2]	-	-	Reserved	-
[1:0]	VLCD	R/W	VLCD voltage configuration also refers to VLCD setting. Supports Real chip GPL813P01X only. Must use OTP test VLCD	VLCD [1:0]: 00:2.8V 01:3.0V 10:3.3V 11:VDD

Support EV chip GPL813V00A only

P_LCD_VLCD_Ctrl			0x0019		LCD Control Register			
Bit	7	6	5	4	3	2	1	0
Function	-				VLCD			
Default	0	0	0	0	1	0	0	1

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	VLCD	R/W	VLCD [3:0] :	1/2 Bias 1/3 Bias 1/4 Bias 1/5 Bias
			0000	2.0V 3.0V 4.0V 5.0V
			0001	2.066 3.099 4.132 5.165
			0010	2.134 3.201 4.268 5.335
			0011	2.2 3.3 4.4 5.5
			0100	2.266 3.399 4.532 5.665
			0101	2.334 3.501 4.668 5.835
			0110	2.4 3.6 4.8 6
			0111	2.466 3.699 4.932 N/A
			1000	2.534 3.801 5.068 N/A
			1001	2.6 3.9 5.2 N/A
			1010	2.666 3.999 5.332 N/A
			1011	2.734 4.101 5.468 N/A

Bit	Function	Type	Description			Condition	
			1100	2.8	4.2	5.6	N/A
			1101	2.866	4.299	5.732	N/A
			1110	2.934	4.401	5.868	N/A
			1111	3	4.5	6	N/A

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9 SLEEP / WAKE UP

9.1 Introduction

GPL813P01X features a sleep mode for power savings. In sleep mode, system clock will stop, and RAM and IO state will retain at their previous states. The sleep mode includes halt mode and standby mode. In the halt mode, system Clock will be stopped, but CLK32768 will keep operating. In the standby mode, both CPU clock and CLK32768 are stopped. Programmer can write the **P_SYSTEM_Ctrl** control register any data to enter sleep mode and further set bit7=1 of P_CLK_32768_En to enter standby mode.

When GPL813P01X wakes up from sleep mode, CPU will run the next instruction from where the system entered sleep. And wakeup will not issue system reset, so any initial configuration needs to be set by software.

Standby mode wake-up source: only key (PortE) wake-up.

Halt mode wake-up source: Key change, TimeBaseA, 128Hz, TimeBaseB, Timer0, Timer1, overflow wake-up (not related to Fosc).

Note.

1. When waking up from standby mode, CPU will wait certain period of time for VLCD voltage becoming stable. After VLCD voltage is stable, CPU starts running. Please reference **POWER ON/WAKE UP STABLE TIME** chapter for more details.

9.2 \$0008, P_WAKEUP_Ctrl

P_WAKEUP_Ctrl		0x0008				Wake-up Source Control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	Timer1	Timer0	TMBB	T128Hz	TMBA	KEYCH
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6]	-	-	Reserved	-
5	Timer1	R/W	Timer1 wakeup. Only CLK32K and TMBB as Timer1 clock source support wakeup function. Because in halt mode, system clock will be disable.	Write: 0: disabled/clear Timer1 wakeup 1: Timer1 wakeup enabled Read: 0: No wakeup 1: Wakeup
4	Timer0	R/W	Timer0 wakeup. Only CLK32K, EXTI and TMBB as Timer0 clock source support wakeup function. Because in halt mode, system clock will be disable.	Write: 0: disabled/clear Timer0 wakeup 1: Timer0 wake-up enabled Read: 0: No wakeup 1: Wakeup
3	TMBB	R/W	TimeBaseB wakeup. The TimeBaseB wake-up frequency depends on	Write: 0: disabled/clear TimeBaseB

Bit	Function	Type	Description	Condition
			register 0x000A[2:0] In real chip(GPL813P01X) not support TimerBase Wake up 0x000A="111": 4096Hz	wakeup 1: TimeBaseB wakeup enabled Read: 0: No wakeup 1: Wakeup
2	T128Hz	R/W	128Hz wakeup. This bit can set wake-up source from 128Hz.	Write: 0: disabled/clear 128Hz wakeup 1: 128Hz wakeup enabled Read: 0: No wakeup 1: Wakeup
1	TMBA	R/W	TimeBaseA wakeup. The TimeBaseA used 1 or 2 Hz depends on register 0x000A.b7	Write: 0: disabled/clear TimeBaseA wakeup 1: TimeBaseA wakeup enabled Read: 0: No wakeup 1: Wakeup
0	KEYCH	R/W	Key change wakeup.	Write: 0: disabled/clear key change wakeup 1: key change wakeup enabled Read: 0: No wakeup 1: Wakeup

9.3 \$0009, P_SYSTEM_Ctrl

P_SYSTEM_Ctrl			0x0009			System Control Register		
Bit	7	6	5	4	3	2	1	0
Function	SLEEP							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	SLEEP	W	Writing any data to this register, CPU will enter sleep mode immediately to save power. Thus, system clock will stop automatically.	Write any data to enter sleep mode.

9.4 Procedure to Sleep

1. LCD set off enters to Halt mode.

LDA #00H

```

STA      P_LCD_Ctrl      ; LCD display disable
LDA      $$XXXX0000
STA      P_IO_Ctrl       ; set PE as general I/O
NOP
LDA      P_IO_PortE_DataLatch ; Latch current Port E data

```

L_SetSleepPort:

```

LDA      #00H
STA      P_WAKEUP_Ctrl    ; Clear the wakeup status
LDA      #%%00000001     ; Set wake source as key change
                                ; user must modify it if necessary

STA      P_WAKEUP_Ctrl    ; Set wake up mask
STA      P_SYSTEM_Ctrl    ; Stop the CPU clock
NOP                                ; these three NOPs are prepared
NOP                                ; for wakeup procedure to avoid
NOP                                ; error instruction process
JMP      Reset            ; Use this instruction to make
                                ; software reset(recommend)

```

2. LCD set off enters to Stand-by mode. (Turn off clock 32768 by setting \$0C=80H)

...

```

LDA      P_IO_PortE_DataLatch ; Latch current Port E data

```

L_SetSleepPort:

```

LDA      #00H
STA      P_WAKEUP_Ctrl    ; Clear the wakeup status
LDA      #%%00000001     ; Set wake source as key change
                                ; user must modify it if necessary

STA      P_WAKEUP_Ctrl    ; Set wake up mask
LDA      #80H
STA      P_CLK_32768_En   ; Turn off clock 32768
STA      P_SYSTEM_Ctrl    ; Stop the CPU clock
NOP                                ; these three NOPs are prepared
NOP                                ; for wakeup procedure to avoid
NOP                                ; error instruction process
JMP      Reset            ; Use this instruction to make
                                ; software reset(recommend)

```

Note:

1. GPL813P01X does not generate hardware reset when system wakes up. It means program will continue to execute the next instruction from where the system enters sleep.

9.5 \$000C, P_CLK_32768_En

P_CLK_32768_En			0x000C			32768Hz Clock Control Register		
Bit	7	6	5	4	3	2	1	0
Function	LFENB	-	-	-	-	Weak	Strong	CAP_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	LFENB	W	This bit is used to control 32K ON/OFF. Halt Mode: This bit must be set to "0". Standby Mode: This bit must be set to "1".	0: 32K clock source enabled 1: 32K clock source disabled
[6:3]	-	-	Reserved	-
[2]	Weak	W	X'tal32K select Normal mode or Weak mode. When Power on release, X'tal32K at Strong mode will keep 1 sec, then enter the Normal mode or Weak mode. X'tal32K Start-up sorted by intensity. Strong>Normal>Weak	0: Normal mode enabled 1: Weak mode enabled
[1]	Strong	W	X'tal32K Strong mode always Disable/ Enable	0: Strong mode disabled 1: Strong mode enabled
[0]	CAP_EN	W	X'tal32K XIN/XOUT without Internal CAP or with Internal CAP.	0: Without Internal CAP 1: With Internal CAP

Note: CAP_EN selects "1" : with Internal CAP, the Crystal 32768 suggest C_L selects 12.5p only.

10 RFC FUNCTION

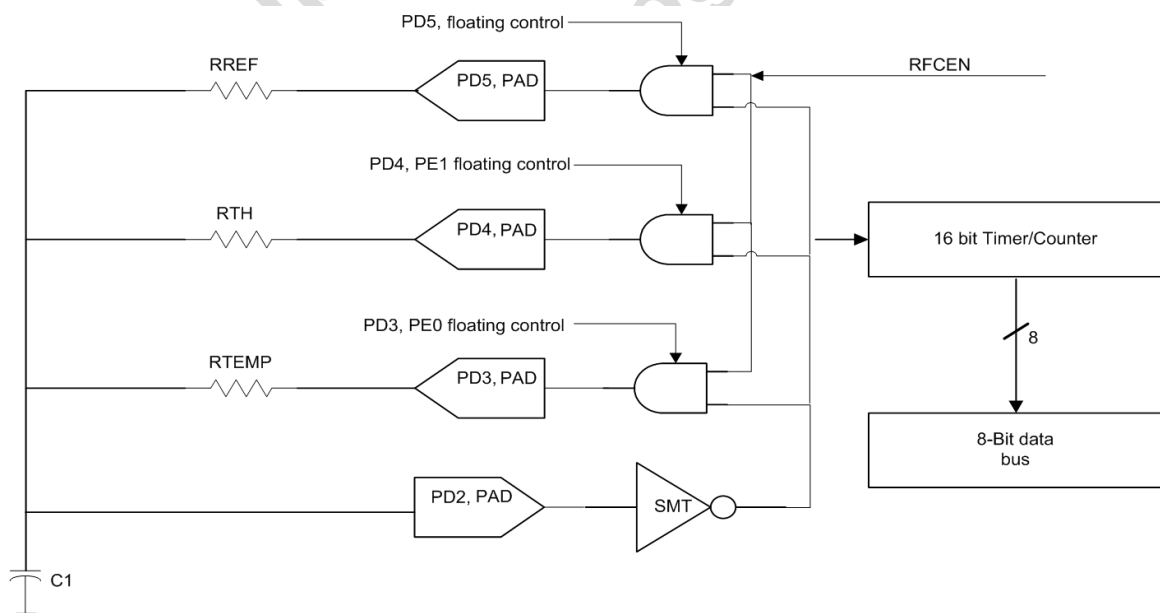
10.1 Introduction

GPL813P01X features a Resistor to Frequency Converter (RFC) function. Because the heat-resistance sensor and moisture-resistance sensor changes its resistance due the change of temperature and humidity, RC time constant will also be changed. Programmers can use this identity to measure temperature or humidity.

10.2 \$0017, P_RFC_Ctrl

P_RFC_Ctrl		0x0017				RFC control register		
Bit	7	6	5	4	3	2	1	0
Function	EXT1EDG	EXT2EDG	-	-	-	-	-	RFCEN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	EXT1EDG	W	PD2(EXT1) interrupt edge	0: Falling edge 1: Rising edge
6	EXT2EDG	W	PD6(EXT2) interrupt edge	0: Falling edge 1: Rising edge
[5:1]	-	Reserved	-	-
0	RFCEN	W	This bit sets RFC function enabled or disabled	0: RFC function disabled 1: RFC function enabled



In RFC application, PD[5:3] used as pass-through (output) pin and connected to sensor. PD2 used as input-floating pin and connected to sensor & capacitor.

10.3 How to use RFC Function

(1). Generate RC pulse:

Step 1: Set Timer0 clock as EXT1 of group A and clear Timer0 count

Step 2: Set one of PD3 / PD4 / PD5 ports to be passed-through (output) for each measurement; the other two ports are PAD-floating.

Step 3: Set PD2 to Input floating.

Step 4: Enable RFC function (\$0017register) to generate RC-Oscillation pulse and enable Timer to record RC count.

10.4 Example Code

Here, assuming PD5 channel is used to perform RFC function

```
LDA    #D_TimerSrcB_VDD+D_TimerSrcA_EXT    ;set timer0 clock as EXT1(PD2)
STA    P_TIMER_Timer0_Ctrl
LDA    #00                                ;clear timer count
STA    P_TIMER_TM0Data_LB
STA    P_TIMER_TM0Data_HB
STA    P_TIMER_TM0Load_Ctrl

LDA    #(Bit4+Bit3)                        ;set PD3/PD4 as output floating
STA    P_IO_PortD_Attrib                    ;set PD5 as output low
STA    RB_IO_PortD_Attrib                  ;set PD2 as input floating
LDA    #Bit5+Bit4+Bit3
STA    P_IO_PortD_Dir
STA    RB_IO_PortD_Dir
LDA    #Bit2
STA    P_IO_PortD_Data
STA    RB_IO_PortD_Data

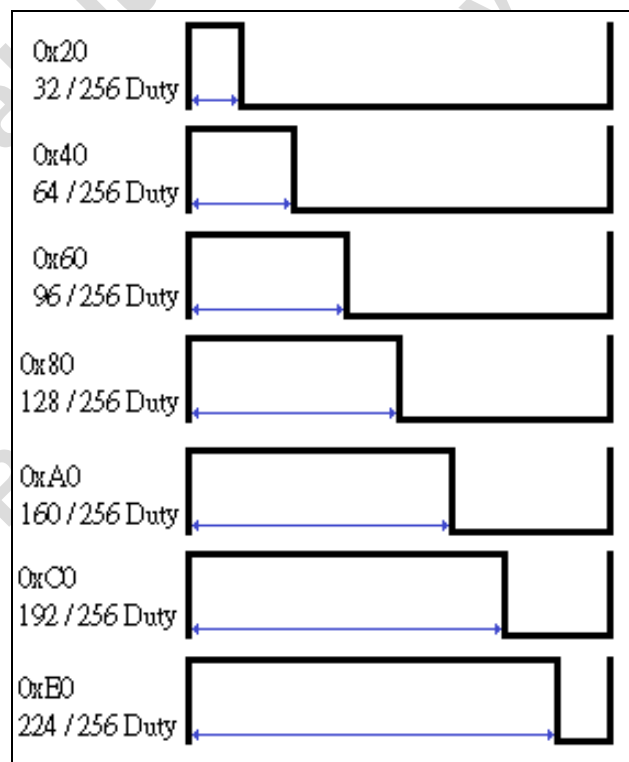
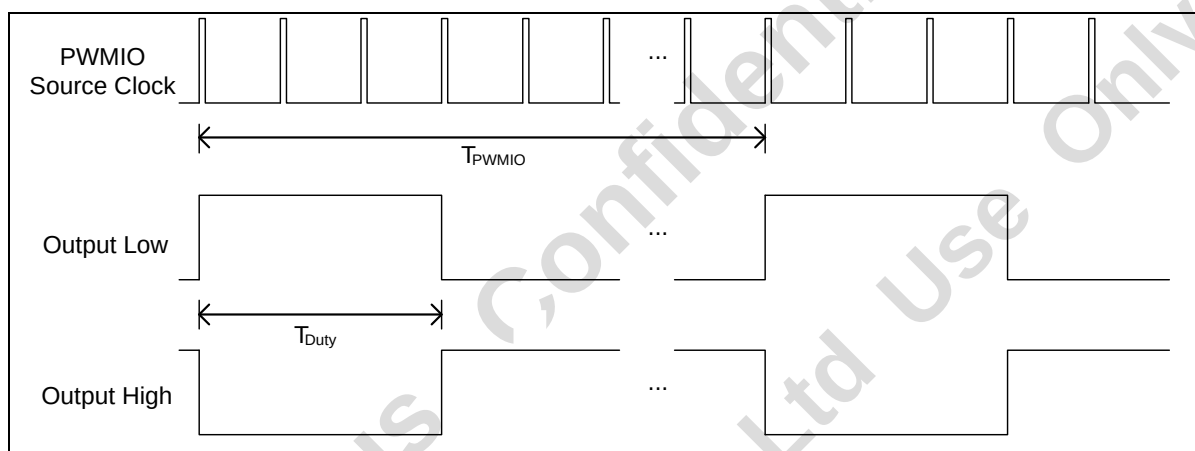
LDA    #D_RfcEn                            ;enable RFC and Timer
STA    P_RFC_Ctrl
LDA    P_TIMER_Timer0_Ctrl
ORA    #D_TimerEn
STA    P_TIMER_Timer0_Ctrl
```


11 PWMIO

11.1 General Description

GPL813P01X supports four PWM IOs sharing IOA[3:0] for LED control, with 256-level duty selection. Programmer can set IOA Output Low or High to decide the duty is high plus or low plus start-up. Please refer to the following Timing Diagram for more information about the relationship between IOA Output low and Output high.

11.2 Timing Diagram



PWM Duty:0x3032~0x3035	T _{Duty}
0x00	NA Duty
0x20	32/256 Duty
0x40	64/256 Duty
0x60	96/256 Duty
0x80	128/256 Duty
0xA0	160/256 Duty
0xC0	192/256 Duty
0xE0	224/256 Duty
0xFF	255/256 Duty

11.3 PWMIO Pin Configuration

Name	I/O	Description
PWMIO3	O	PWMIO3 (shared with IOA3)
PWMIO2	O	PWMIO2 (shared with IOA2)
PWMIO1	O	PWMIO1 (shared with IOA1)
PWMIO0	O	PWMIO0 (shared with IOA0)

11.4 Register Summary

Name	Address	Description
P_PWMIO_Ctrl	0x3030	PWMIO control register
P_PWMIO_IO0_DUTY	0x3032	PWMIO0 duty selection register
P_PWMIO_IO1_DUTY	0x3033	PWMIO1 duty selection register
P_PWMIO_IO2_DUTY	0x3034	PWMIO2 duty selection register
P_PWMIO_IO3_DUTY	0x3035	PWMIO3 duty selection register

11.5 Register Definition

P_PWMIO_Ctrl	0x3030	PWMIO Control Register						
Bit	7	6	5	4	3	2	1	0
Function	-	Clear	PWMIO3EN	PWMIO2EN	PWMIO1EN	PWMIO0EN	CLKSEL	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	-	-	Reserved	-
6	Clear	W	Clear PWM IO timer	1 = Clear
5	PWMIO3EN	R/W	PWM I/O3 Enabled/Disabled	0 = Disabled 1 = Enabled
3	PWMIO2EN	R/W	PWM I/O2 Enabled/Disabled	0 = Disabled 1 = Enabled

Bit	Function	Type	Description	Condition
2	PWMIO1EN	R/W	PWM I/O1 Enabled/Disabled	0 = Disabled 1 = Enabled
1	PWMIO0EN	R/W	PWM I/O0 Enabled/Disabled	0 = Disabled 1 = Enabled
0	CLKSEL	R/W	Set the clock source of PWM I/O	00 = Fosc 01 = Fosc/Div 10 = Timer0 overflow 11 = Timer1 overflow

Note:

- When clock source is Timer TMO overflow, timer data ranges in between 0000H~FFFEH; FFFFH is not allowed.
- PWMIO output status change timing will be synchronized with CPU clock. Thus, CPU clock frequency should be faster than PWMIO clock frequency.

P_PWMIO_IO0_DUTY		0x3032			PWMIO0 Duty Control			
Bit	7	6	5	4	3	2	1	0
Function	DUTY							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	DUTY	R/W	Duty Control for PWMIO0	Write = Duty control Read = Represent duty

P_PWMIO_IO1_DUTY		0x3033			PWMIO1 Duty Control			
Bit	7	6	5	4	3	2	1	0
Function	DUTY							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	DUTY	R/W	Duty Control for PWMIO1	Write = Duty control Read = Represent duty

P_PWMIO_IO2_DUTY		0x3034			PWMIO2 Duty Control			
Bit	7	6	5	4	3	2	1	0
Function	DUTY							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	DUTY	R/W	Duty Control for PWMIO2	Write = Duty control Read = Represent duty

P_PWMIO_IO3_DUTY		0x3035		PWMIO3 Duty Control				
Bit	7	6	5	4	3	2	1	0
Function	DUTY							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	DUTY	R/W	Duty Control for PWMIO3	Write = Duty control Read = Represent duty

12 RESET

12.1 General Description

Total of three reset sources are available in GPL813P01X: POR(power on reset), LVR(low voltage reset), and WDTR(watchdog reset),

Reset type	Condition	Enable / Disable Control	Result
POR (Power on reset)	Power On	Always enabled	CPU reset, all registers will be cleared to default value.
LVR (Low voltage reset)	VDD drops below LVR level	EV: Code Option Real Chip: Code Option	
WDTR (watchdog reset)	User does not clear WDT in 1 second.	EV: Code Option Real Chip: Code Option	

Note: Watchdog Enable / Disable for EV/Real chip is code option;

13 VOLTAGE DETECTOR/LOW VOLTAGE RESET

13.1 General Description

LVD

There are four programmable detection levels provided. Program can be written to monitor the P_LVD_Ctrl for low voltage event detection. Programmers can also check register P_LVD_Status.bit0 to get LVD status, reading this bit will get low voltage detection status. The LVD function is Code Option setting to Enable or Disable.

Note:

The LVD status (0x001B.b0) will be updated every 16ms after LVD function and 32K clock enable. So the programmer cannot read LVD status immediately after LVD function and 32K clock enable.

LVR

The low voltage reset (LVR) module is Code Option setting.

13.2 \$001B , P_LVD_Status

P_LVD_Status			0x001B			Low voltage detector control register		
Bit	7	6	5	4	3	2	1	0
Function	LVDLEV1	LVDLEV0	-					LVDS
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6]	LVDLEV[1:0]	R/W	Low voltage detection level selection	00→2.2V 01→2.4V 10→2.7V 11→3.3V
[5:1]	-	-	Reserved	-
0	LVDS	R	Read this bit to acquire LVD status After enable LVD function, user need delay 16ms time at least then can read this status.	LVDS : 0 VDDSW > LVD level; 1 VDDSW < LVD level.

Note: When P_LVD_Status.b0 is read "1" and if attempt to read P_LVD_Status again, system will retain the read value ("1") unless VDD is higher than LVD level that makes system change the value of P_LVD_Status.b0 back to "0".

13.3 Restrictions of LVD Interrupt

P_INT_Ctrl.b6: enable LVD interrupt and notice the condition to trigger LVD interrupt.

First, VDD must be higher than LVD level. When VDD drops lower than LVD level, LVD interrupt occurs once until VDD rises higher than LVD level, and if VDD later drops lower than LVD again, it will retrigger LVD interrupt again. If VDD voltage is continuously keeping lower than LVD level, LVD interrupt will not retrigger again.

LVD Interrupt must set disabled when entering halt mode. LVD Interrupt can be set enabled in Normal Mode.

Note: When entering the halt mode, LVD Interrupt must be disabled. When waking up from halt mode, set LVD Interrupt enabled to detect LVD level.

	Normal Mode	Halt Mode	Standby Mode
CPU Clock	ON	OFF	OFF
32768Hz	ON or OFF	ON	OFF

13.4 Procedure LVD interrupt to enter halt mode

For Example:

```

LDA      # D_TMBAInt+D_TMBBInt+D_Timer0Int+D_LvdInt
STA      P_INT_Ctrl
CLI
...
...
LDA      # D_TMBAInt+D_TMBBInt+D_Timer0Int
STA      P_INT_Ctrl           ; LVD Interrupt must set Disable when enter the halt mode.
NOP
LDA      P_IO_PortE_DataLatch ; Latch current Port E data

```

L_SetSleepPort:

```

LDA      #00H
STA      P_WAKEUP_Ctrl        ; Clear the wakeup status
LDA      #%00000011          ; Set wake source as 2Hz + key change
                                ; user must modify it if needed
STA      P_WAKEUP_Ctrl        ; Set wakeup mask
STA      P_SYSTEM_Ctrl        ; Stop CPU clock
NOP      ; these three NOPs are prepared
NOP      ; for wakeup procedure to avoid
NOP      ; error instruction process
LDA      # D_TMBAInt+D_TMBBInt+D_Timer0Int+D_LvdInt
STA      P_INT_Ctrl           ; After Halt mode wakeup to set LVD Interrupt Enable to detect LVD.
...

```

14 HARDWARE SUMMARY

14.1 GPL813P01X Mapping

\$0000-\$007F	Control Registers
\$0080-\$00FF	SRAM
\$0100-\$017F	Reserved
\$0180-\$01FF	SRAM (stack)
\$3000-\$3035	Control Registers2
\$3036-\$3DFF	Reserved
\$3E00-\$3E73	DPRAM
\$3E74-\$BFFF	Reserved
\$C000-\$FFDF	User OTP ROM
\$FFE0-\$FFF9	GP used reserved
\$FFFA-\$FFFF	User Interrupt Vector

GPL813P01X Control Register

Adress	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
\$00	P_IO_PortA_Dir	00h	W	-	-	-	-	PA3	PA2	PA1	PA0
				PortA Direction control: 0: IN; 1:OUT							
\$01	P_IO_PortC_Dir	00h	W	PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
				Port B Direction control: 0: IN; 1:OUT							
\$02	P_IO_PortA_Data	xxh	R	PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0
		00h	W	PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0
\$03	P_IO_PortE_Data	xxh	R	PE7	PE6	PE5	PE4	PE3	PE2	PE1	PE0
		00h	W	PE7	PE6	PE5	PE4	PE3	PE2	PE1	PE0
\$04	P_IO_PortE_DataLatch	xxh	R	Read this port to latch the data of Port E							
\$05	P_IO_PortC_Data	XXH	R	PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
		00H	W	PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
\$06	P_IO_Ctrl	00H	W	PA[3:2]	PA[1:0]	PC[7:4]	PC[3:0]	PE[7:4]	PE[3:2]	PE1	PE0
				Attribute Control: 1: Special function I/O, 0: General I/O							
\$07	P_LCD_Ctrl	00h	W	Duty1	Duty0	Bias0	Bias1	Duty2	Duty3	-	LCD En
				Duty[3:0]: 6=>1/8duty; 7=>1/6duty; 8=>1/5duty; 9=>1/4duty; A=>1/3duty; Bias[1:0]: 0=> 1/4bias; 1=>1/3bias; 2=>1/2bias; LCD En: 0=>Disable; 1=>Enable.							
\$08	P_WAKEUP_Ctrl	00h	R	-	TIMER2	TIMER1	TIMER0	TMBB	T128Hz	TMBA	KEYCH
		00h	W	-	TIMER2	TIMER1	TIMER0	TMBB	T128Hz	TMBA	KEYCH
				TMBB depends on \$000A.[2:0], TMBA depends on \$000A.[7]							

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
\$09	P_SYSTEM_Ctrl	xxh	W	Writing any data into \$0009 to enter sleep mode							
\$0A	P_TIMER_TimerX_Ctrl	xxh	R	R 0.5Hz	R 1Hz	R 2Hz	R 4Hz	R 8Hz	R 16Hz	R 32Hz	R 64Hz
		00h	W	TMBA: 0: 2Hz 1: 1Hz	-	-	-	-	TMBBSEL[2:0]: 000: 4Hz, 001: 8Hz 010: 16Hz, 011: 32Hz 100: 64Hz, 101: 128Hz 110: 1024Hz 111: 4096Hz		
\$0B	P_IO_PortE_Dir	00h	W	PE7	PE6	PE5	PE4	PE3	PE2	PE1	PE0
				Port E Direction control: 0: IN; 1: OUT							
\$0C	P_CLK_32768_En	00h	W	LFENB	-	-	-	-	Weak	Strong	CAP EN
				LFENB: 32KHz Frequency oscillator enable, low active Weak: X'tal32K weak mode and Normal mode Select. Strong: X'tal32K strong mode always turn on Enable/Disable CAP_EN: X'tal32K Internal XIN/XOUT with CAP or Without CAP.							
\$0D	P_INT_Ctrl	00h	R	EXT2INT	LVDINT	TM0INT	TMBBINT	TMBAINT	128HzINT	2KHzINT	EXT1INT
		00h	W	EXT2EN	LVDEN	TM0EN	TMBBEN	TMBAEN	128HzEN	2KHzEN	EXT1EN
\$0E	P_Reset_Status	00h	R/W	-	-	-	-	-	-	FlagON	FlagC
				FlagON : Reset due to ACON key in at Stop mode(Stop all clock source) FlagC : Reset due to ACON key in at Normal mode(include only CPU sleep)							
\$0F	P_WDT_Clear		W	Writing any data into \$000F to clean watchdog							
\$10	P_IO_PortB_Dir	00h	W	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
				Port B Direction control: 0: IN; 1: OUT							
\$11	P_IO_PortD_Dir	00h	W	PD7	-	PD5	PD4	PD3	PD2	PD1	PD0
				Port D Direction control: 0: IN; 1: OUT EXT1EDG: 0 : PD2(EXT1) falling edge interrupt; 1 : PD2(EXT1) rising edge interrupt							
\$12	P_IO_PortB_Data	00h	R	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
			W	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
\$13	P_IO_PortD_Data	00h	R	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0
			W	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0
\$14	P_IO_PortB_Attrib	00h	W	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
				Attribute Control: 1: Special function I/O, 0: General I/O							
\$15	P_IO_PortD_Attrib	00h	W	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0
				Attribute Control: 1: Special function I/O, 0: General I/O							
\$16	P_AUDIO_BZ_Ctrl	00h	W	C32KOUT	-	-	-	-	-	Buzzer2	Buzzer1
				C32KOUT 0 : PD7 as normal IO ; 1 : PD7 output CLK32K Buzzer[2:1] 11: PortD[1:0] as buzzer output 00: PortD[1:0] as normal output							
\$17	P_RFC_Ctrl	00h	W	EXT1EDG	EXT2EDG						RFCEN

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
				RFCEN : 0 : RFC function disable, 1 : RFC function enable EXT1EDG : 0 : PD2(EXT1) falling edge interrupt, 1 : PD2(EXT1) rising edge interrupt EXT2EDG : 0 : PD6(EXT2) falling edge interrupt, 1 : PD6(EXT2) rising edge interrupt							
\$18	P_CLK_CPU_Ctrl	00h	R/W	SYSCLK2	SYSCLK1	SYSCLK0	-	-	CPUCK2	CPUCK1	CPUCK0
				0 0 0: Fosc=225KHz 0 1 0: Fosc=500KHz 1 0 0: Fosc=1800KHz 1 1 0: Fosc=1000KHz X X 1: Fosc=4000KHz GPL813V00A used only. X X 1: Fosc=4000KHz GPL813P01X used only.				-	-	0 0 0; F _{CPU} =Fosc 0 0 1; F _{CPU} =Fosc / 2 0 1 0; F _{CPU} =Fosc / 4 0 1 1; F _{CPU} =Fosc / 8 1 x x; F _{CPU} =Fosc / 16	
\$19	P_LCD_VLCD_Ctrl	00h	R/W	-	-	GREEN1	GREEN0	-	-	VLCD1	VLCD0
				-	-	Halt green[5:4] 00: Reserved 01: Reserved 10: Halt current min. 11: Halt current Nor.		-	-	VLCD [1:0] :00:2.8V 01:3.0V 10:3.3V 11:VDD	
\$1A	P_CLK_HFDIV_Ctrl	00h	R/W	-	-	-	-	-	-	DIV1	DIV0
				-	-	-	-	-	-	0 0; Fosc=Fosc/2 0 1; Fosc=Fosc/4 1 0; Fosc=Fosc/8 1 1; Fosc=Fosc/16	
\$1B	P_LVD_Status	00h	R/W	LVDLEV1	LVDLEV0	-	-	-	-	-	LVDS
				LVDLEV[1:0] : 00 : 2.2V; 01 : 2.4V; 10 : 2.7V; 11 : 3.3V LVDS (read only) : 0 : VDDSW > LVD level; 1 : VDDSW < LVD level							
\$20	P_TIMER_TM0Data_LB	Xxh	R	B7	B6	B5	B4	B3	B2	B1	B0
			W	B7	B6	B5	B4	B3	B2	B1	B0
\$21	P_TIMER_TM0Data_HB	Xxh	R	B7	B6	B5	B4	B3	B2	B1	B0
			W	B7	B6	B5	B4	B3	B2	B1	B0
\$22	P_TIMER_Timer0_Ctrl	00h	R/W	TMR0EN	-	EXTA	-	TCKB1	TCKB0	TCKA1	TCKA0
				TMR0EN : 0→Timer disable, 1→Timer enable; EXTA : 0→EXTA=EXT1 as timer clock, 1→EXTA=EXT1/2 as timer clock; TCKB[1:0] : 00→CLK32K; 01→TMBB; 10→EXTA; 11→VDD TCKA[1:0] : 00→Fosc; 01→Fosc_DIV; 10→EXT1; 11→VDD							
\$23	P_TIMER_TM0Load_Ctrl		W	Write \$23H to reload Timer0 register value							
\$25	P_TIMER_TM1Data	Xxh	R	B7	B6	B5	B4	B3	B2	B1	B0
			W	B7	B6	B5	B4	B3	B2	B1	B0
\$26	P_TIMER_Timer1_Ctrl	00h	R/W	TMR1EN	-	-	-	-	-	TM1CK1	TM1CK0

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
				TMR1EN : 0→Timer1 disable; 1→Timer1 enable TM1CK[1:0] : 00→ CLK32K; 01→TMBB; 10→ RCHF; 11→ HFDIV							
\$27	P_TIMER_TM1Load_Ctrl	Xxh	W	Write \$27H to reload Timer1 register value							
\$2D	P_INT_Ctrl2	00h	R	-	-	-	-	-	-	-	TM1INT
			W	-	-	-	-	-	-	-	TM1INT
\$40	P_INT_Ext1_Clear	Xxh	W	Writing any data into \$40 will clear EXT1INT							
\$41	P_INT_2KHz_Clear	Xxh	W	Writing any data into \$41 will clear 2KINT							
\$42	P_INT_128Hz_Clear	Xxh	W	Writing any data into \$42 will clear 128INT							
\$43	P_INT_TimeBaseA_Clear	Xxh	W	Writing any data into \$43 will clear 2HZINT							
\$44	P_INT_TimeBaseB_Clear	Xxh	W	Writing any data into \$44 will clear TMXINT							
\$45	P_INT_Timer0_Clear	Xxh	W	Writing any data into \$45 will clear TM0INT							
\$46	P_INT_LVD_Clear	Xxh	W	Writing any data into \$46 will clear LVDINT							
\$48	P_INT_Timer1_Clear	Xxh	W	Writing any data into \$48 will clear TM1INT							
\$3030	P_PWMIO_Ctrl	00h	R/W	-	Clear	PWMIO3EN	PWMIO2EN	PWMIO1EN	PWMIO0EN	CLKSEL	-
\$3032	P_PWMIO_IO0_DUTY	00h	R/W	DUTY							
\$3033	P_PWMIO_IO1_DUTY	00h	R/W	DUTY							
\$3034	P_PWMIO_IO2_DUTY	00h	R/W	DUTY							
\$3035	P_PWMIO_IO3_DUTY	00h	R/W	DUTY							

Note: If the register bits are not defined, it will not be implemented in the real chip; its readout value should be random. However, it is defined as 0 in this table for simplification.

15 APPENDIX A. INSTRUCTION SET

No.	Syntax	Code No.	Code Size	CPU Cycle	Addressing Mode
001	ADC #dd	69H	2	2	immediate
002	ADC aa	65H	2	3	zero page
003	ADC aa,X	75H	2	4	zero page indexed x
004	ADC aaaa	6DH	3	4	absolute
005	ADC aaaa,X	7DH	3	4	absolute indexed x
006	ADC aaaa,Y	79H	3	4	absolute indexed y
007	ADC (aa,X)	61H	2	6	indexed indirect x
008	ADC (aa),Y	71H	2	5	indirect indexed y
009	AND #dd	29H	2	2	immediate
010	AND aa	25H	2	3	zero page
011	AND aa,X	35H	2	4	zero page indexed x
012	AND aaaa	2DH	3	4	absolute
013	AND aaaa,X	3DH	3	4	absolute indexed x
014	AND aaaa,Y	39H	3	4	absolute indexed y
015	AND (aa,X)	21H	2	6	indexed indirect x
016	AND (aa),Y	31H	2	5	indirect indexed y
017	ASL A	0AH	1	2	accumulator
018	ASL aa	06H	2	5	zero page
019	ASL aa,X	16H	2	6	zero page indexed x
020	ASL aaaa	0EH	3	6	absolute
021	ASL aaaa,X	1EH	3	7	absolute indexed x
022	BCC ??	90H	2	2/3/4*	relative
023	BCS ??	B0H	2	2/3/4*	relative
024	BEQ ??	F0H	2	2/3/4*	relative
025	BIT aa	24H	2	3	zero page
026	BIT aaaa	2CH	3	4	absolute
027	BMI ??	30H	2	2/3/4*	relative
028	BNE ??	D0H	2	2/3/4*	relative
029	BPL ??	10H	2	2/3/4*	relative
030	BRK	00H	1	7	implied
031	BVC ??	50H	2	2/3/4*	relative
032	BVS ??	70H	2	2/3/4*	relative
033	CLC	18H	1	2	implied
034	CLD	D8H	1	2	implied
035	CLI	58H	1	2	implied
036	CLV	B8H	1	2	implied
037	CMP #dd	C9H	2	2	immediate
038	CMP aa	C5H	2	3	zero page
039	CMP aa,X	D5H	2	4	zero page indexed x
040	CMP aaaa	CDH	3	4	absolute
041	CMP aaaa,X	DDH	3	4	absolute indexed x
042	CMP aaaa,Y	D9H	3	4	absolute indexed y
043	CMP (aa,X)	C1H	2	6	indexed indirect x
044	CMP (aa),Y	D1H	2	5	indirect indexed y
045	CPX #dd	E0H	2	2	immediate
046	CPX aa	E4H	2	3	zero page
047	CPX aaaa	ECH	3	4	absolute
048	CPY #dd	C0H	2	2	immediate
049	CPY aa	C4H	2	3	zero page
050	CPY aaaa	CCH	3	4	absolute
051	DEC aa	C6H	2	5	zero page
052	DEC aa,X	D6H	2	6	zero page indexed x
053	DEC aaaa	CEH	3	6	absolute
054	DEC aaaa,X	DEH	3	7	absolute indexed x
055	DEX	CAH	1	2	implied
056	DEY	88H	1	2	implied
057	EOR #dd	49H	2	2	immediate
058	EOR aa	45H	2	2	zero page

No.	Syntax	Code No.	Code Size	CPU Cycle	Addressing Mode
059	EOR aa,X	55H	2	2	zero page indexed x
060	EOR aaaa	4DH	3	3	absolute
061	EOR aaaa,X	5DH	3	3	absolute indexed x
062	EOR aaaa,Y	59H	3	3	absolute indexed y
063	EOR (aa,X)	41H	2	2	indexed indirect x
064	EOR (aa),Y	51H	2	2	indirect indexed y
065	INC aa	E6H	2	5	zero page
066	INC aa,X	F6H	2	6	zero page indexed x
067	INC aaaa	EEH	3	6	absolute
068	INC aaaa,X	FEH	3	7	absolute indexed x
069	INX	E8H	1	2	implied
070	INY	C8H	1	2	implied
071	JMP aaaa	4CH	3	3	absolute
072	JMP (aaaa)	6CH	3	5	indirect absolute
073	JSR aaaa	20H	3	6	absolute
074	LDA #dd	A9H	2	2	immediate
075	LDA aa	A5H	2	3	zero page
076	LDA aa,X	B5H	2	4	zero page indexed x
077	LDA aaaa	ADH	3	4	absolute
078	LDA aaaa,X	BDH	3	4	absolute indexed x
079	LDA aaaa,Y	B9H	3	4	absolute indexed y
080	LDA (aa,X)	A1H	2	6	indexed indirect x
081	LDA (aa),Y	B1H	2	5	indirect indexed y
082	LDX #dd	A2H	2	2	immediate
083	LDX aa	A6H	2	3	zero page
084	LDX aa,Y	B6H	2	4	zero page indexed y
085	LDX aaaa	AEH	3	4	absolute
086	LDX aaaa,Y	BEH	3	4	absolute indexed y
087	LDY #dd	A0H	2	2	immediate
088	LDY aa	A4H	2	3	zero page
089	LDY aa,X	B4H	2	4	zero page indexed x
090	LDY aaaa	ACH	3	4	absolute
091	LDY aaaa,X	BCH	3	4	absolute indexed x
092	LSR A	4AH	1	2	accumulator
093	LSR aa	46H	2	5	zero page
094	LSR aa,X	56H	2	6	zero page indexed x
095	LSR aaaa	4EH	3	6	absolute
096	LSR aaaa,X	5EH	3	7	absolute indexed x
097	NOP	EAH	1	2	implied
098	ORA #dd	09H	2	2	immediate
099	ORA aa	05H	2	3	zero page
100	ORA aa,X	15H	2	4	zero page indexed x
101	ORA aaaa	0DH	3	4	absolute
102	ORA aaaa,X	1DH	3	4	absolute indexed x
103	ORA aaaa,Y	19H	3	4	absolute indexed y
104	ORA (aa,X)	01H	2	6	indexed indirect x
105	ORA (aa),Y	11H	2	5	indirect indexed y
106	PHA	48H	1	3	implied
107	PHP	08H	1	3	implied
108	PLA	68H	1	4	implied
109	PLP	28H	1	4	implied
110	ROL A	2AH	1	2	accumulator
111	ROL aa	26H	2	5	zero page
112	ROL aa,X	36H	2	6	zero page indexed x
113	ROL aaaa	2EH	3	6	absolute
114	ROL aaaa,X	3EH	3	7	absolute indexed x
115	ROR A	6AH	1	2	accumulator
116	ROR aa	66H	2	5	zero page
117	ROR aa,X	76H	2	6	zero page indexed x
118	ROR aaaa	6EH	3	6	absolute
119	ROR aaaa,X	7EH	3	7	absolute indexed x
120	RTI	40H	1	6	implied
121	RTS	60H	1	6	implied

No.	Syntax	Code No.	Code Size	CPU Cycle	Addressing Mode
122	SBC #dd	E9H	2	2	immediate
123	SBC aa	E5H	2	3	zero page
124	SBC aa,X	F5H	2	4	zero page indexed x
125	SBC aaaa	EDH	3	4	absolute
126	SBC aaaa,X	FDH	3	4	absolute indexed x
127	SBC aaaa,Y	F9H	3	4	absolute indexed y
128	SBC (aa,X)	E1H	2	6	indexed indirect x
129	SBC (aa),Y	F1H	2	5	indirect indexed y
130	SEC	38H	1	2	implied
131	SED	F8H	1	2	implied
132	SEI	78H	1	2	implied
133	STA aa	85H	2	3	zero page
134	STA aa,X	95H	2	4	zero page indexed x
135	STA aaaa	8DH	3	4	absolute
136	STA aaaa,X	9DH	3	5	absolute indexed x
137	STA aaaa,Y	99H	3	5	absolute indexed y
138	STA (aa,X)	81H	2	6	indexed indirect x
139	STA (aa),Y	91H	2	6	indirect indexed y
140	STX aa	86H	2	3	zero page
141	STX aa,Y	96H	2	4	zero page indexed y
142	STX aaaa	8EH	3	4	absolute
143	STY aa	84H	2	3	zero page
144	STY aa,X	94H	2	4	zero page indexed x
145	STY aaaa	8CH	3	4	absolute
146	TAX	AAH	1	2	implied
147	TAY	A8H	1	2	implied
148	TSX	BAH	1	2	implied
149	TXA	8AH	1	2	implied
150	TXS	9AH	1	2	implied
151	TYA	98H	1	2	implied

* If no branch jump, the CPU cycle is 2. If branch jump condition is confirmed and jump to the same page, the CPU cycle is 3. If jump to different page, the CPU cycle is 4.